



MIS

Identify the measures to improve cyber security with example.

Improving cybersecurity involves implementing comprehensive defense mechanisms, often called **controls** or **countermeasures**, that protect an organization's information and information systems (IS) from unauthorized access, disruption, or destruction. Since organizations face diverse threats, they typically utilize layers of controls, known as **defense-in-depth**.

The most fundamental measure is adopting a proactive **risk management** strategy, which aims to identify, control, and minimize the impact of threats to reduce risk to acceptable levels.

Here is a detailed breakdown of cybersecurity measures (controls) and examples, suitable for an examination answer:

1. Foundational Measures: Risk Mitigation and Education

Before implementing specific technologies, organizations address the probability and impact of risks through mitigation strategies.

Measure	Definition and Example
Risk Mitigation Strategies	Concrete actions taken against identified risks.
Risk Acceptance	Accepting the potential risk, continuing operations with no controls, and absorbing any resulting damages. <i>Example: A homeowner deciding not to insure their home, thus accepting the financial risk of damage.</i>

Measure	Definition and Example
Risk Limitation	Implementing controls to minimize the impact if a threat occurs. <i>Example: A homeowner limiting risk by installing an alarm system.</i>
Risk Transference	Transferring the risk by using other means to compensate for the loss, typically by purchasing insurance. <i>Example: The vast majority of homeowners purchasing insurance on their houses and possessions.</i>
User Education and Training	Considered the single most valuable control, ensuring every member of the organization is aware of the vital importance of information security.

2. Major Types of Information Security Controls

Organizations implement three major types of controls to protect the components of an information system (data, software, hardware, and networks).

A. Physical Controls

Physical controls prevent unauthorized individuals from gaining access to a company's facilities.

- **Examples:** Walls, doors, fencing, gates, locks, badges, guards, and alarm systems. More sophisticated examples include pressure sensors, temperature sensors, and motion detectors.

B. Access Controls

Access controls restrict unauthorized individuals from using information resources. This involves **Authentication** (determining identity) and **Authorization** (determining permissions).

Authentication Factor	Description and Example
Something the User Is (Biometrics)	Examines innate physical characteristics. <i>Example: Fingerprint scans, retina scans, iris recognition, or facial recognition.</i>
Something the User Has	Includes physical devices. <i>Example: Regular ID cards, smart ID cards (with an embedded chip), or tokens (with embedded chips and a digital display showing a login number).</i>
Something the User Does	Authentication based on behavior. <i>Example: Voice recognition (matching a spoken phrase) or signature recognition (matching</i>

Authentication Factor	Description and Example
	<i>speed and pressure of a signature).</i>
Something the User Knows	Authentication via secrets, typically passwords and passphrases. <i>Example: A strong password should be long, include uppercase/lowercase letters, numbers, and special characters, and not be recognizable words. Example of a passphrase converted to a strong password: "9GaMmD/" (from "go ahead make my day")*.</i>
Multifactor Authentication	Implementing more than one type of authentication, such as a password plus a biometric scan, to increase reliability and security.

C. Communications Controls (Network Controls)

Communications controls secure the movement of data across networks.

Control Type	Function and Example
Firewalls	A system (hardware, software, or both) that prevents a specific type of information from moving between untrusted networks (like the Internet) and private networks. <i>Example: An organization may use an External Firewall facing the Internet and an Internal Firewall facing the company network, separated by a Demilitarized Zone (DMZ), where servers handling web requests are located.</i>
Anti-Malware Systems	Software packages, also called antivirus (AV) software, that identify and eliminate malicious software. <i>Example: Norton AntiVirus, McAfee VirusScan, and Trend Micro PC-cillin.</i>
Whitelisting	A process where a company identifies and permits acceptable software to run on its computers, preventing all unapproved software from launching.
Blacklisting	Identifies software or websites that are not allowed to run or be accessed in the company environment. <i>Example: A company blacklisting peer-to-peer file sharing on its systems.</i>
Encryption	The process of converting an original message into an unreadable form. <i>Example: Secure Socket Layer (SSL) is an encryption standard used for secure transactions like credit card purchases and online banking.</i>

Control Type	Function and Example
Virtual Private Networks (VPNs)	Private networks that use a public network (usually the Internet) to securely connect users using encryption. <i>Example: VPNs use tunneling to encrypt each data packet and place it inside another packet for confidential transmission across the Internet.</i>
Employee Monitoring Systems	Systems that monitor employees' computers, email, and Internet surfing activity to protect networks against internal mistakes and threats. <i>Example: Using software like SpectorSoft or Websense to identify employees visiting questionable websites or wasting time.</i>

3. Resilience Measures: Business Continuity Planning

Business continuity planning (also known as a disaster recovery plan) provides guidance to keep the business operating after a disaster, ensuring critical business functions continue and restoring operations quickly.

- **Hot Site:** A fully configured computer facility that duplicates the company's computing resources (including applications and communication links), providing near-real-time recovery.
- **Warm Site:** Provides many hot site services but often lacks the company's specific applications or user workstations.
- **Cold Site:** Provides only rudimentary facilities (like a building with heating, AC, and humidity control) but no computer hardware.

Analyze the impact of BI on Decision making.

The impact of Business Intelligence (BI) on decision-making is profound, fundamentally transforming how managers, knowledge workers, and organizations formulate strategies and respond to pressures. BI systems enable comprehensive data analysis, providing timely insights necessary for solving complex problems and achieving organizational goals.

1. Definition and Scope of Business Intelligence

Business Intelligence (BI) is defined as a broad category of applications, technologies, and processes for **gathering, storing, accessing, and analyzing data to help business users make better decisions.**

The ultimate goal of BI systems is to ensure that information systems provide the **right information to the right person, in the right amount, at the right time, in the right format.** BI achieves this by using data, typically sourced from **data warehouses and data marts**, to provide actionable insights.

Historically, BI primarily supported management decision-making, but today its scope is expanding, making applications available to front-line personnel (like call center operators), customers, suppliers, and regulators.

2. Context: Why Decision-Makers Need BI Support

Decision-making involves a systematic process composed of the intelligence, design, and choice phases. Managers require BI support because modern decision-making is challenging due to several trends:

- **Increasing Alternatives:** Innovations and global markets constantly increase the number of alternatives available, requiring sophisticated computer assistance for searching and comparing options.
- **Time Pressure:** Most decisions must be made under **time pressure**, making it difficult to manually process information quickly enough. BI addresses this by providing **near-real-time information** to managers.
- **Complexity and Uncertainty:** Increased uncertainty in the business environment makes decisions more complex, demanding sophisticated analysis.
- **Access to Remote Information:** Decision-makers often need to rapidly access remote information, consult with experts, or conduct group sessions across different locations.

BI systems are particularly valuable for providing computer-based support for **complex, nonroutine decisions**, primarily benefiting middle managers and knowledge workers.

3. Direct Impact on Decision Outcomes

BI supports the quality and efficiency of the decision process across all managerial decision types, including structured, semi-structured, and unstructured problems in operational control, management control, and strategic planning.

- **Improved Agility and Speed:** ERP systems often integrate organizational resources, providing essential information on business performance across functional areas, which **significantly improves managers' ability to make better, more timely decisions**. This allows organizations to react quickly to changing business conditions.
- **Prediction and Forecasting:** When properly analyzed, Big Data leveraged through BI tools can **reveal valuable patterns and information that were previously hidden**. At its core, BI is used to generate **predictions** by applying mathematics to huge quantities of data to infer probabilities (e.g., the likelihood an email is spam or forecasting fraud).
- **Fundamental Transformation:** At the highest level, BI can be used to **fundamentally transform the ways a company competes in the marketplace**, supporting entirely new business models and strategies.

4. BI Applications for Decision Analysis and Presentation

Decision support is enabled by specific BI applications that facilitate analysis and help present results clearly.

A. Applications for Data Analysis

1. **Multidimensional Analysis (Online Analytical Processing or OLAP):** This capability involves "slicing and dicing" data stored in a dimensional format, **drilling down** to greater detail, and **aggregating** the data. This approach allows users to analyze data intuitively from the perspective of **business dimensions** (such as product, time period, and geographic area).
2. **Data Mining:** This process searches for valuable business information in large databases or data warehouses. While general BI shows *what* has happened, data mining helps explain *why* it is happening and **predicts what will happen in the future**.

- *Example:* Data mining can be used in **retailing and sales** to predict sales or prevent theft; in **banking** to forecast bad loans; and in **policework** (predictive policing) to predict when and where future crimes are most likely to occur.
3. **Decision Support Systems (DSSs):** These systems combine **models and data** to analyze semi-structured and unstructured problems with extensive user involvement. DSSs enable the user to conduct complex hypothetical assessments:
- **Sensitivity Analysis:** Examines the impact that changes in input variables (like reorder points or inflation rates) have on other parts of the decision model.
 - **What-If Analysis:** Attempts to predict the impact on the solution based on a change in assumptions (input data).
 - **Goal-Seeking Analysis:** A "backward" approach that calculates the necessary input values required to achieve a desired level of output (e.g., what sales volume is needed to reach a specific profit target).

B. Applications for Presenting Results

1. **Dashboards (Digital Dashboards):** These are a special form of IS that support all managers by providing **rapid access to timely information** and direct access to management reports. They are user-friendly, supported by graphics, and enable managers to examine exception reports and **drill down** into detailed data. They often display **Key Performance Indicators (KPIs)**, which are critical measures for business success.
- *Example:* The Management Cockpit concept uses an elaborate set of dashboards displayed on the walls of a meeting room to help managers visualize how different business factors interrelate.
2. **Geographic Information Systems (GIS):** These systems capture, integrate, manipulate, and display data using digitized maps. The graphical format makes it easy for managers to **visualize the data** for planning, problem-solving, and decision-making by linking every record to a geographical location (geocoding).

3. **Corporate Performance Management (CPM)**: This BI area monitors and manages an organization's performance based on **Key Performance Indicators (KPIs)** such as revenue, ROI, and costs. CPM helps align raw materials, human resources, and production schedules with customer demand, thus reducing business risk.

Explain Data warehouse and Data Mart in an organization.

Data Warehouse

A **Data Warehouse** is defined as a repository of historical data that is organized by subject to support decision-makers in the organization.

Purpose and Context

The primary function of a Data Warehouse is to support **Business Intelligence (BI)** applications. BI is a broad category of applications, technologies, and processes designed for gathering, storing, accessing, and analyzing data to help business users make **better decisions**.

Historically, organizations used their operational databases to process transactions (Online Transaction Processing or OLTP). However, operational databases are not optimized for complex analysis for the following reasons:

- They are optimized for transactional processing (accessing a single record at a time), whereas decision-makers need to access **large groups of related records**.
- Complicated queries in operational databases can take a long time and **degrade the performance** of the database systems.
- Operational databases are designed to be continuously updated, requiring extra processing; Data Warehouses are typically **read-only**, eliminating this overhead.

A Data Warehouse solves these problems by providing users with access to corporate data that is specifically organized and optimized for analysis.

Key Characteristics of Data Warehouses

Data Warehouses and the smaller Data Marts share several characteristics that distinguish them from operational databases:

1. **Organized by Business Dimension or Subject:** Data are organized by subjects meaningful to the business, such as customer, vendor, product, or region, instead of being organized by business process (like inventory control or order entry).
2. **Time Variant:** They maintain **historical data** (data that includes time as a variable), often storing years of data. This historical perspective is critical for identifying and analyzing trends and deviations.
3. **Nonvolatile:** Users **cannot change or update the data** in a Data Warehouse, meaning the stored data accurately reflects history. Updates occur through controlled IT load processes, not by end users.
4. **Integrated:** Data are collected from **multiple systems** (internal and external) and integrated around common identifiers (like a customer ID) to create a comprehensive view.
5. **Multidimensional:** Data are stored in a **multidimensional structure**, often represented by a **data cube**. This allows users to view and analyze data intuitively from the perspective of multiple business dimensions (e.g., product, geographic area, and time period).
6. **Use Online Analytical Processing (OLAP):** Data Warehouses are designed to support decision makers using **OLAP**, which involves the analysis of accumulated data through "slicing and dicing" and "drilling down" to detail.

Data Warehouses are primarily used by **large companies** because they are very expensive to build and maintain.

Data Mart

A **Data Mart** is a **low-cost, scaled-down version of a data warehouse** that is designed to meet the end-user needs of a specific strategic business unit (SBU)

or an individual department.

Key Characteristics and Differences from Data Warehouses

- **Size and Scope:** Data Marts are much smaller than enterprise-wide data warehouses. They store data for a single application or a few applications, such as marketing or finance, rather than supporting the needs of the entire organization.
- **Cost and Implementation:** They are typically **less expensive** and can be implemented much **more quickly** than a full data warehouse, often in less than 90 days.
- **Control:** Data Marts often support local rather than central control, placing power with the user group.
- **Use Case:** Groups that require specialized analysis or a single BI application often need only a Data Mart.

An organization's BI infrastructure might utilize **independent Data Marts** (which may contain inconsistent data definitions and source systems) or a **hub and spoke** architecture, where a central Data Warehouse feeds data into multiple dependent Data Marts. When Data Marts source their data from the central repository, the data in these marts still comprise the "**single version of the truth**" for decision-support purposes.

What is Data Mart and Data Warehouses? Give two examples which show generation of Big Data

Data Warehouse

A **Data Warehouse** is a repository of **historical data** that is organized by subject to **support decision-makers** in the organization. It serves as the foundation for **Business Intelligence (BI) systems**, which are applications, technologies, and

processes for accessing and analyzing data to help business users make better decisions.

Data Warehouses possess several key characteristics that distinguish them from typical operational databases:

- **Organization:** They are organized by **business dimension or subject** (e.g., customer, vendor, product, or region), rather than by business process (like inventory control).
- **Time Variant:** They maintain **historical data** (data that includes time as a variable), allowing organizations to analyze deviations, trends, and long-term relationships, unlike transactional systems which maintain only recent data.
- **Nonvolatile:** Users cannot change or update the data; thus, the warehouse reflects history accurately. Updates occur through IT-controlled load processes.
- **Integrated:** Data are collected from multiple internal and external systems and then integrated around common subjects (e.g., a customer identifier) to create a comprehensive, consolidated view.
- **Multidimensional:** They typically store data in a **multidimensional structure**, often represented by a **data cube**. This structure organizes data by business dimensions such as product, geographic area, and time period, allowing for intuitive analysis.
- **Use of OLAP:** They support **Online Analytical Processing (OLAP)**, which involves the analysis of accumulated data through operations like "slicing and dicing" or "drilling down" to detail.

Due to the expense and effort required to build and maintain them, Data Warehouses are used primarily by **large companies**.

Data Mart

A **Data Mart** is a **low-cost, scaled-down version of a data warehouse**.

- **Scope:** It is designed specifically to meet the **end-user needs** of an **individual department** or a **strategic business unit (SBU)**.

- **Implementation:** Data Marts can be implemented **more quickly** than full Data Warehouses, often in **less than 90 days**.
 - **Control:** They support **local rather than central control**, conferring power on the user group.
 - **Use Case:** Typically, user groups that need only a single or a few BI applications require a Data Mart rather than a full Data Warehouse.
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Two Examples Showing the Generation of Big Data

Big Data refers to the **vast and constantly increasing amounts of data** that modern organizations must capture, store, process, and analyze. It is characterized by three distinct features: **volume, velocity, and variety**. The following examples demonstrate the immense volume and high velocity at which digital data is generated today:

Example 1: Social Media Content Generation (Volume and Velocity)

Social media platforms generate tremendous volumes of Big Data every hour and every day:

- **Facebook and Photo Uploads:** Facebook members upload **more than 10 million new photos every hour**. In addition, users click the "like" button or leave a comment **nearly 3 billion times every day**. Even on a daily basis, users upload **more than 350 million digital photos just to Facebook**.
- **Microblogging Sites:** Snapchat users alone share **more than 100 million "snaps" daily**, and the volume of messages on Twitter grows at **200 percent every year**, exceeding 450 million tweets per day by mid-2013.

Example 2: Digital Content Processing (Volume)

Massive volumes of digital content are generated by companies, resulting in a dramatic shift from analog to digital storage:

- **Google's Daily Processing:** Google was processing **more than 24 petabytes of data every day** in 2013. (A petabyte is approximately 1,000 terabytes, or trillions of bytes.)

- **Digital Dominance:** As recently as the year 2000, only 25 percent of the stored information in the world was digital; the rest was analog (stored on paper, film, etc.). By 2013, the amount of stored information in the world was estimated to be around 1,200 exabytes, of which **less than 2 percent was non-digital**. This shows the massive scale of digital data generation.

Explain the steps involved in knowledge capturing.

The process of **knowledge capturing** is one of the essential steps in the **Knowledge Management System (KMS) cycle**.

This cycle involves six dynamic steps, which are refined over time as the environment changes. Knowledge management (KM) is the process that helps organizations manipulate important knowledge that comprises part of the organization's memory, often in an unstructured format.

The full steps involved in the KMS cycle are:

1. **Create Knowledge:** Knowledge is first generated as people **determine new ways of doing things** or **develop know-how**. Sometimes, **external knowledge** is also brought into the organization.
2. **Capture Knowledge:** New knowledge must be **identified as valuable** and subsequently **represented in a reasonable way**.
3. **Refine Knowledge:** The newly captured knowledge must be **placed in context so that it is actionable**. This phase is crucial because it requires **tacit qualities (human insights)** to be captured along with explicit facts.
4. **Store Knowledge:** Once refined, the useful knowledge must be **stored in a reasonable format in a knowledge repository**. This allows others in the organization to access it.

5. **Manage Knowledge:** Similar to managing a library, the stored knowledge must be **kept current**. It needs to be **reviewed regularly to verify that it is relevant and accurate**.
 6. **Disseminate Knowledge:** Finally, the knowledge must be **made available in a useful format** to anyone in the organization who needs it, regardless of their location or time.
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Focusing on Knowledge and Expertise

The ultimate goal of knowledge capture and the KMS is to utilize **knowledge**, which consists of data and/or information that has been organized and processed to convey **understanding, experience, accumulated learning, and expertise** as they apply to a current business problem.

A major challenge in knowledge management is dealing with the two types of knowledge:

- **Explicit Knowledge:** This is the more objective, rational, and technical knowledge that has been **codified (documented)** in a form that can be distributed to others. Historically, management information systems have focused on capturing, storing, managing, and reporting this type of knowledge. *Example:* A firm's human resources policy manual containing a documented description of how to process a job application.
- **Tacit Knowledge:** This is the **cumulative store of subjective or experiential learning**. It includes an organization's experiences, insights, expertise, know-how, trade secrets, skill sets, understanding, and learning. It is highly personal, unstructured, imprecise, and difficult to formalize or transfer. Capturing these **tacit qualities (human insights)** is part of the refining phase of the KMS cycle. *Example:* The specialized knowledge a salesperson possesses about specific customers' needs due to working with them over time.

For organizations to be successful, they need to integrate both explicit and tacit knowledge into formal information systems. KMSs are intended to help an organization cope with rapid change, turnover, and downsizing by making the expertise of its human capital widely accessible.

How Does IT Impact Organizations?

The impact of Information Technology (IT) on organizations is fundamental, wide-ranging, and often disruptive, affecting their profitability, competitive position, daily operations, internal structure, and relationship with external stakeholders. IT is widely considered the **backbone of any business** across all functional areas, including Accounting, Finance, Marketing, Human Resources, Operations Management, and MIS.

The overall effect of IT is to provide the **foundation for modern business enterprises**. This impact can be analyzed across several critical areas:

1. Strategic and Competitive Impact

IT is integral to an organization's ability to compete and survive in the modern business environment.

- **Foundation for Modern Business:** Information systems provide the foundation for modern business enterprises. To compete successfully in a global, rapidly changing, intensely competitive, 24/7/365, and information-intensive environment, organizations **must use IT effectively**.
- **Driving Competitive Advantage:** Organizations use IT and information systems to **increase their profitability, gain market share, and improve customer service**. IT is vital in achieving **competitive advantage** in measures such as cost, quality, or speed. Any information system that helps an organization either achieve a competitive advantage or reduce a competitive disadvantage is considered a **Strategic Information System (SIS)**.
- **Business Transformation:** At the highest level, IT can be used to **fundamentally transform the ways a company competes in the marketplace**, supporting entirely new business models and strategies. For example, the use of IT is essential to achieving **operational effectiveness** ("I can do the same thing more efficiently than you can").
- **Disruption:** IT and software are **disrupting every industry**, forcing companies to become **more agile** and respond to competitive threats more quickly, efficiently, and effectively. Examples of software disruption include the fate of

Borders bookstore following Amazon's online rise, and the transformation of financial services where nearly every transaction is software-driven.

2. Operational and Functional Impact

IT dramatically improves the efficiency and effectiveness of core business processes.

- **Efficient Operations:** IT enables organizations to **perform business processes more efficiently and effectively**. Information systems collect, process, store, analyze, and disseminate information for specific purposes.
- **Cross-Functional Integration:** Systems like **Enterprise Resource Planning (ERP) systems** tightly integrate information from different functional areas (Accounting, Finance, Marketing, etc.) via a common database, which **greatly increases organizational productivity** and enhances communication across the firm. IT also facilitates **cross-functional business processes** which originate in one department and end in another.
- **Core Capabilities (as summarized in the major capabilities of IS):** IT enables organizations to perform high-speed, high-volume numerical computations; provide fast, accurate communication and collaboration; store huge amounts of information; allow quick and inexpensive access to vast amounts of global information; interpret vast amounts of data quickly; and automate both semiautomatic business processes and manual tasks.
- **Specific Functional Support:** IT supports specific activities across the business:
 - **Accounting/Finance:** Forecasting revenues, determining best sources and uses of funds, and conducting audits.
 - **Marketing/Sales:** Developing new goods and services (product analysis), determining facility location (site analysis), and managing customer relationships.
 - **Production/Operations:** Processing customer orders, developing production schedules, and managing inventory.
 - **Human Resources:** Managing the recruiting process, screening applicants, and administering compensation/benefits.

3. Impact on Organizational Structure and Employees

IT alters the nature of managerial and nonmanagerial work.

- **Reduction of Middle Managers:** IT makes managers more productive and can increase the number of employees reporting to a single manager. This can **decrease the number of managers and experts**, potentially leading to fewer managerial levels and fewer staff and line managers.
- **Changes to the Manager's Job:** IT provides managers with **near-real-time information**, which means they have less time to make decisions, potentially making their jobs more stressful. IT also provides tools like **dashboards** and business analytics to help them manage information volumes. Managers will increasingly supervise **geographically dispersed employees and teams**.
- **Job Elimination vs. Creation:** Relentless cost-cutting can lead to organizations doing "more with less," and IT sometimes replaces people with machines, leading to layoffs. However, IT also creates entirely **new categories of jobs** (e.g., electronic medical record keeping).
- **Employee Experience:** Computerization can cause people to experience a **loss of identity**, feeling like "just another number". Conversely, IT allows employees **flexibility** (e.g., telecommuting) in their work schedule and location.

4. Risk, Misuse, and Security Impact

IT introduces new vulnerabilities that organizations must manage.

- **Misuse and Failures:** While IT systems present many examples of success, there are numerous examples of **IS failures** that illustrate lessons learned, as the misuse of IS can be very expensive.
- **Vulnerability:** IT resources face increasing vulnerability due to factors such as the **interconnected, wirelessly networked business environment**, the smaller/faster/cheaper nature of computers and storage, and a general lack of management support for security.
- **Security Measures:** Organizations must implement **information security controls** (defense-in-depth) to protect their information and systems.

Furthermore, IT facilitates **e-discovery applications** that can extract relevant concepts and deduce patterns of behavior in legal contexts.

Describe the tools that augment the traditional SDLC.

The traditional **Systems Development Life Cycle (SDLC)** is a structured, sequential method used for large-scale IT projects. Various alternative methodologies and tools augment this traditional approach, either by providing faster delivery, automating steps, or accommodating shifting user requirements. These augmentations include alternative development methods and specific development tools.

Alternative Development Methods

These methodologies replace or streamline the sequential phases of the traditional SDLC, focusing on speed, collaboration, and flexibility.

1. **Joint Application Design (JAD):** This is a group-based tool used for collecting user requirements and creating system designs, primarily within the systems analysis and design stages of the SDLC. JAD involves a meeting where analysts and all relevant users jointly define and agree on the system requirements, which saves a tremendous amount of time in the overall process.
2. **Rapid Application Development (RAD):** RAD is a systems development method designed to rapidly produce a high-quality system. It often incorporates other tools like JAD and prototyping. The development process in RAD is **iterative**, meaning that requirements, designs, and the system itself undergo a sequence of continuous improvements. Importantly, RAD produces the **functional components of a final system**, rather than just models. RAD effectively compresses the design and testing phases.
3. **Agile Development:** This methodology delivers system **functionality in rapid iterations**, often measured in weeks. Agile development emphasizes frequent communication, development, testing, and delivery. The goal is to maximize

the team's ability to deliver quickly and respond effectively to emerging user requirements, focusing on doing only what is immediately necessary for success.

4. **End-User Development:** This approach allows the organization's **end users to develop their own applications** with minimal or no formal assistance from the IT department. The user maintains control of the application and can change it as needed, ensuring that the system directly meets user requirements and increases user acceptance.

Specific Tools Augmenting Development

These are discrete tools or frameworks designed to enhance efficiency, reusability, or clarity during the development process:

1. **Prototyping:** This approach begins by defining an initial list of user requirements, then constructing a **small-scale working model, or prototype**, of the system. The system is then refined through several **iterations** based on continuous user feedback. Prototyping is typically used to develop user interfaces, and if the system proves viable, the developers can use the prototype as the basis for building the full system.
2. **Integrated Computer-Assisted Software Engineering (ICASE) Tools:** CASE tools automate many tasks in the SDLC. **Upper CASE tools** automate the early stages (investigation, analysis, and design), while **Lower CASE tools** automate the later stages (programming, testing, and maintenance). **ICASE tools** specifically provide the links between these upper and lower tools.
3. **Component-Based Development:** This is a software development methodology that constructs applications using **standard, reusable components**. These components are generally reusable applications that fulfill a single, specific function, such as a **shopping cart**, a user authentication module, or a **catalog**.
4. **Object-Oriented (OO) Development:** This methodology focuses on **modeling real-world entities, or "objects,"** rather than the tasks to be performed. In an OO system, developers identify objects—which represent real-world entities like a customer or a course—each possessing **properties** (data values) and

operations (behaviors) that can be performed on those properties. This approach allows for the reuse of existing objects

Analyze how social computing is inspiring customer service?

social computing is fundamentally transforming customer service by creating a new, highly empowered environment where customers have high expectations and businesses must respond quickly and appropriately. This transformation is driven by the capabilities of social computing to facilitate communication, collaboration, and rapid feedback.

Social computing, a type of IT that combines social behavior and information systems, creates value by focusing on improving collaboration and interaction among people and encouraging user-generated content.

Here is an analysis of how social computing is inspiring (and demanding) improvements in customer service:

1. Customer Empowerment and Elevated Expectations

Social computing has made customers **incredibly empowered**. This empowerment stems from the fact that customers now routinely use social media (such as Twitter and Facebook) to inform the world how they feel about companies, their products, and their services.

- **Public Accountability:** Customers are mindful that they can air their complaints in public, creating pressure on businesses to resolve issues quickly. Companies are closely monitoring social computing because they are aware of the negative comments that can be posted by social network members.
- **Choosing Interaction:** Empowered customers choose how they interact with companies and brands and have **elevated expectations** concerning their experiences. They are involved with businesses not just as purchasers, but also as **advocates and influencers**.

- **Utilizing Crowds:** Customers know how to use the **wisdom and power of crowds and communities** to their benefit. This collective power can force policy changes, as seen when a Facebook campaign organized by a customer successfully persuaded Qantas airlines to change its policy regarding musical instruments in cargo holds.

2. Strategic Opportunities for Businesses

Companies are shifting their focus to view social computing not just as a vulnerability, but as an **opportunity** to proactively involve customers and improve service.

- **Rapid Response and Problem Alleviation:** Social computing requires businesses to **respond to customers quickly and appropriately**. By identifying problems quickly, businesses can alleviate customer anger. For instance, after a customer tweeted displeasure with Citibank's long hold times, a social media agent immediately tweeted back and resolved the issue, turning the disgruntled customer into an advocate.
- **Turning Disgruntled Customers into Champions:** Social computing provides opportunities for businesses to **turn disgruntled customers into champions for the firm**. For example, P.F. Chang's used social media as an easy opportunity to make the most of a customer's experience, which resulted in the customer praising the restaurant for caring about its customers.
- **Real-Time Engagement and Support:** Companies can use social networking sites to provide real-time customer service. Best Buy, for example, used its Twitter account (@twelpforce) to allow its Geek Squad members and corporate employees to staff the account, providing answers to customer questions. This strategy provided real-time customer service by utilizing the company's existing human capital.
- **Integration with Customer Service Systems:** Social media platforms can be integrated directly with customer service software. Rosetta Stone, for example, integrated its Facebook page with customer service software to monitor Wall posts, flag those requiring a company response, and ensure prompt answers.

3. Impact on Customer Relationship Management (CRM)

Social computing has dramatically altered the practice of **Customer Relationship Management (CRM)**. CRM systems now rely heavily on social media data to provide a comprehensive view of the customer.

- **Collaborative CRM:** Social computing applications like blogs and wikis are important to companies that value customer input into their product and service offerings, as well as into new product development. **Collaborative CRM systems** enable customers to provide direct feedback to the organization.
- **Customer Touch Points:** Social media platforms (like e-mail, Web sites, and smartphones) are now recognized as essential **customer touch points** that must be managed by organizational CRM systems.
- **Gaining Insights:** Companies gain first-hand insight into their customers' desires by listening to conversations on social networks. This continuous feedback contributes to a "**single version of the truth**" about the customer and is vital for maximizing customer satisfaction.

Categorize the approach to managing information across an entire organization

Data Governance is the approach to managing information across an entire organization. It is a formal set of business processes and policies designed to ensure that data are handled in a certain, well-defined fashion across the entire organization. The goal of this approach is to follow unambiguous rules for creating, collecting, handling, and protecting the organization's information.

The objective of data governance is to make information:

- **Available.**
- **Transparent.**
- **Useful** for the people who are authorized to access it.

This management extends from the moment data enters the organization until it is outdated and deleted.

Core Strategies and Implementations for Managing Information

Managing information resources across an entire organization involves several integrated systems and concepts:

1. **Master Data Management (MDM):** This is one specific strategy for implementing data governance. **MDM** is a process that spans all organizational business processes and applications. It aims to provide the company with the ability to store, maintain, exchange, and synchronize a **consistent, accurate, and timely "single version of the truth"** for the company's **master data**. Master data are core data, such as customer, product, employee, vendor, or geographic location, that span the enterprise information systems.
2. **Information Systems (IS) Management:** In a modern organization, the responsibility for managing information resources is **divided between the MIS department and the end users**. This is due to the evolution of the management information systems (MIS) function, where computers are now located in all departments and almost all employees use them in their work, a situation known as **end user computing**.
 - Generally speaking, the **MIS department is responsible for corporate-level and shared resources**.
 - The **end users are responsible for departmental resources**.
 - Regardless of the division of labor, the MIS department and the end users must work in **close cooperation**.
3. **IT Infrastructure and Governance:** The technological approach for managing information enterprise-wide requires establishing a structured IT infrastructure:
 - The foundation of this infrastructure is the **information technology platform**, formed by the **IT components** (hardware, software, networks, and databases).
 - **IT personnel** use these components to develop information systems, oversee security and risk, and manage data; these activities are called **information technology services**.
 - The IT components plus IT services comprise the organization's **information technology infrastructure**.

4. **Strategic Planning and Alignment:** Organizational-wide information management is guided by comprehensive planning:
- The IT resources needed are identified in the **IT strategic plan**, which must be **aligned with the organization's strategic plan**.
 - The IT strategic plan sets long-range goals for the IT infrastructure and major IT initiatives needed to achieve the organization's goals.
 - An **IT steering committee**, comprised of managers and staff from various units, is set up to establish IT priorities and ensure the MIS function meets the needs of the enterprise.

Explain the impact of information system on organization and society.

The impact of Information Systems (IS) and Information Technology (IT) is **wide-ranging, global, and disruptive**, affecting organizations and society by serving as the **foundation for modern business enterprises**.

1. Impact on Organizations

IT affects profitability, operations, competitive positioning, and internal structure.

A. Strategic and Competitive Effects

- IT is essential for organizational survival and for achieving a **competitive advantage** in areas like cost, quality, and speed.
- Any system that grants a competitive edge or reduces a disadvantage is a **Strategic Information System (SIS)**.
- IT is disrupting nearly every industry, forcing companies to become **more agile** and respond to competitive threats quickly and effectively.

B. Operational and Functional Effects

- IS collects, processes, stores, analyzes, and disseminates information to execute business processes **more efficiently and effectively**.

- IT supports high-volume computations, stores massive amounts of data, and enables rapid, accurate communication and collaboration.
- IT facilitates **cross-functional business processes** and, through systems like Enterprise Resource Planning (ERP), integrates information across functional areas (e.g., Accounting, Marketing, HR) via a common database, **increasing organizational productivity**.

C. Structural and Managerial Effects

- IT can **reduce the number of middle managers** by making managers more productive and increasing the number of employees reporting to a single supervisor.
- IT provides managers with **real-time or near-real-time information**, which can sometimes increase job stress by requiring faster decision-making.
- IT supports new flexible work arrangements like **telecommuting**.
- The **misuse of IS** can be expensive, and system failures highlight the need for careful management.

2. Impact on Society

IT introduces profound changes to human life, connectivity, and well-being.

A. Positive Societal Effects

- IT makes the world **smaller**, enabling more people to communicate, collaborate, and compete, thereby **leveling the digital playing field**.
- IT provides **flexibility in work** (e.g., working from anywhere, anytime).
- IT enables **improvements in healthcare** through administrative systems, fraud detection, and by making useful medical information widely available.
- IT provides **opportunities for people with disabilities** (e.g., through specialized input technologies).
- Robots (intelligent systems applications) are becoming more common and are expected to **take over mundane chores**.

B. Negative Societal Effects

- The **digital divide** refers to the wide gap between those who have access to information technology and those who do not.
- Prolonged IT use can cause **health problems** for individuals, such as job stress and **repetitive strain injuries** (e.g., carpal tunnel syndrome from keyboards).
- IT can place employees on "**constant call**," meaning they are never truly away from the office.
- The vast amount of medical information available on the Web can potentially **misinform patients** about their health problems, leading researchers to caution against self-diagnosis.

Write note on e-commerce. How is E-commerce supported by MIS? Give one case to describe the same.

Note on E-Commerce

Electronic commerce (EC or e-commerce) describes the process of **buying, selling, transferring, or exchanging products, services, or information via computer networks, including the Internet**. E-commerce systems are a type of **interorganizational information system (IOS)** that connect two or more organizations.

E-commerce is often confused with **electronic business (e-business)**, which is a broader concept. E-business includes not only the buying and selling of goods and services but also servicing customers, collaborating with business partners, and performing electronic transactions *within* an organization.

The impact of e-commerce is profound:

- **Increased Reach and Market Share:** E-commerce increases an organization's **reach** (the number of potential customers to whom the company can market its products). It provides unparalleled opportunities for companies to expand worldwide at a small cost, increase market share, and reduce costs.

- **Entrepreneurship:** E-commerce has removed many of the barriers that previously impeded entrepreneurs seeking to start their own businesses.
- **Types:** E-commerce can be categorized based on the parties involved, including **Business-to-Consumer (B2C) electronic commerce**, where organizations sell to individuals, and **Business-to-Business (B2B) electronic commerce**, where both parties are businesses. Other types include Consumer-to-Consumer (C2C), Business-to-Employee (B2E), E-government (G2C and G2B), and Mobile commerce (m-commerce).
- **Mechanisms:** Transactions are conducted using mechanisms such as **electronic catalogs, electronic auctions (forward and reverse), e-storefronts, e-malls, and e-marketplaces**.

How E-Commerce is Supported by MIS

E-commerce is critically dependent on the **Management Information Systems (MIS) function** because MIS personnel provide the necessary technical foundation for all electronic transactions.

1. **Providing IT Infrastructure:** The MIS function is responsible for **providing the information technology infrastructure necessary for electronic commerce to function**. This infrastructure includes the company's essential components like **networks, intranets, and extranets**. These systems allow information to flow seamlessly throughout the enterprise and between organizations, supporting interorganizational operations.
2. **Ensuring Security:** The MIS function is primarily responsible for ensuring that **electronic commerce transactions are secure**. This involves implementing the necessary network controls, such as **firewalls, encryption, and anti-malware systems**, to protect the information transmitted during online buying and selling.
3. **Consultative Partnership:** While the MIS department manages shared and corporate-level IT resources, the responsibility for effective e-commerce use is shared in a partnership with end users. The end user is primarily responsible for effectively using the Internet and electronic commerce in their business, and they **work with the MIS department to accomplish this task**. The MIS

department acts as an advisor, helping to incorporate electronic commerce into the business.

4. **Supporting Functional Areas:** The MIS function directly supports all functional areas. For example, MIS personnel develop applications that enable the firm to do business with its external partners, such as suppliers and customers.

Case to Describe E-Commerce Support by MIS: Macy's Online Presence

The case of Macy's illustrates how the MIS function supports and manages the technical complexity required for a successful e-commerce strategy.

Macy's, a traditional **clicks-and-mortar organization**, dramatically expanded its e-commerce business to make its Web sites (like www.macys.com and www.bloomingdales.com) the primary contact interface for its customers. Its strategy was to provide a **multichannel mechanism** for customers to shop seamlessly in stores, online, and via mobile devices.

To keep its online operations functioning smoothly and efficiently, Macy's implemented specialized software and systems that require ongoing MIS support:

- **Performance Monitoring:** Macy's utilizes **Splunk** software to support its e-commerce initiatives. It also uses **DynaTrace Software** to test the performance of new e-commerce applications they deploy.
- **Real-Time Visibility:** Macy's integrated DynaTrace with an application called **TruSight** (from Coradient). These integrated applications allow company analysts to **examine Web site performance from the perspective of users** by providing real-time visibility into the performance and availability of Web applications.

This case demonstrates MIS support by showing that simply having an e-commerce website is insufficient; the organization must deploy complex, specialized information systems to constantly monitor, maintain, and test the digital delivery channel, ensuring optimal performance and availability for millions of customer

Explain the challenges faced by Knowledge management in different business scenarios.

Knowledge Management (KM) is a process that helps organizations manipulate important knowledge that comprises part of the organization's memory, usually in an unstructured format. **Knowledge management systems (KMSs)** refer to the use of modern information technologies—the Internet, intranets, extranets, databases—to systematize, enhance, and expedite intrafirm and interfirm knowledge management. The KMSs are intended to help an organization cope with turnover, rapid change, and downsizing by making the expertise of the organization's human capital widely accessible.

For an organization to be successful, knowledge, as a form of capital, must exist in a usable form. Knowledge is distinct from data (a collection of facts) and information (organized or processed data); **knowledge is information in action**. KMSs primarily deal with two types of knowledge:

1. **Explicit knowledge:** Deals with more objective, rational, and technical knowledge. This consists of policies, procedural guides, reports, products, strategies, goals, core competencies, and IT infrastructure, and is the knowledge that has been codified (documented) in a form that can be distributed to others or transformed into a process or a strategy.
2. **Tacit knowledge:** The cumulative store of subjective or experiential learning, which is highly personal and hard to formalize. It is generally undocumented.

A functioning KMS follows a cycle that consists of six steps:

1. **Create knowledge:** New knowledge is generated.
2. **Capture knowledge:** New knowledge must be identified as valuable and be represented in a reasonable way.
3. **Refine knowledge:** New knowledge must be placed in context so that it is actionable. This is where tacit qualities (human insights) must be captured along with explicit facts.

4. **Store knowledge:** Useful knowledge must then be stored in a reasonable format in a knowledge repository so that others in the organization can access it.
5. **Manage knowledge:** Like a library, the knowledge must be kept current. It must be reviewed regularly to verify that it is relevant and accurate.
6. **Disseminate knowledge:** Knowledge must be made available in a useful format to anyone in the organization who needs it, anywhere and anytime.

Implementing effective KMSs presents several challenges:

- Employees must be willing to **share their personal tacit knowledge**. Organizations must create a knowledge management culture that rewards employees for contributing their expertise to the knowledge base.
- The organization must continually **maintain and upgrade its knowledge base**. Specifically, it must incorporate new knowledge and delete old, outdated knowledge.
- Companies must be willing to **invest in the resources** needed to carry out these operations.

Identify the five factors that contribute to the increasing vulnerability of information resources, and provide a specific example of each one?

Information security is the protection of an organization's information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. Organizational information resources are subject to myriad threats. A **threat** to an information resource is any danger to which a system may be exposed. The **exposure** of an information resource is the harm, loss, or damage that can result if a threat compromises that resource. An information resource's **vulnerability** is the possibility that the system will be harmed by a threat.

Five key factors are contributing to the increasing vulnerability of organizational information resources, making it much more difficult to secure them:

Factor	Specific Example	Source
Today's interconnected, interdependent, wirelessly networked business environment	Example: The Internet.	
Smaller, faster, cheaper computers and storage devices	Examples: Netbooks, thumb drives, iPads.	
Decreasing skills necessary to be a computer hacker	Example: Information system hacking programs circulating on the Internet.	
International organized crime taking over cybercrime	Example: Organized crime has formed transnational cybercrime cartels. Because it is difficult to know exactly where cyber attacks originate, these cartels are extremely hard to bring to justice.	
Lack of management support	Example: Suppose that your company spent \$10 million on information security countermeasures last year, and they did not experience any successful attacks on their information resources. Short-sighted management might conclude that the company could spend less during the next year and obtain the same results. Bad idea.	

Explain the applications of computer networks.

A **computer network** is a system that connects computers and other devices (e.g., printers) via communications media so that data and information can be transmitted among them. Networks support organizations in various functions.

The two major types of networks are **local area networks (LANs)**, which encompass a limited geographic area and are usually composed of one communications medium, and **wide area networks (WANs)**, which encompass a broad geographical area and are usually composed of multiple communications media. An organization's network, composed of interconnected multiple LANs and WANs, is called an **enterprise network**.

Networks support businesses and other organizations in six major categories of applications:

1. **Discovery:** Involves browsing and information retrieval, and provides users the ability to view information in databases, download it, and/or process it. Discovery tools enable business users to efficiently find needed information. These tools include search engines, directories, and portals.
 - **Search engines** (like Google, Bing, Yahoo!, and Ask) search for specific information by keywords.
 - **Metasearch engines** (like KartOO) search several engines at once and then integrate the findings to answer user queries.
 - **Portals** provide a Web-based, personalized gateway to information and knowledge from different IT systems and the Internet.
2. **Communication:** Provides fast, inexpensive communications, via **e-mail** (the largest-volume application running over the Internet), **call centers**, **chat rooms** (virtual meeting places), and **voice communications** (like VoIP). Communications tools provide business users with a seamless interface among team members, colleagues, business partners, and customers.
3. **Collaboration:** Refers to mutual efforts by two or more entities (individuals, groups, or companies) who work together to accomplish tasks. Collaboration is enabled by **workflow systems** (the movement of information as it flows through work procedures) and **virtual collaboration**.
 - **Work groups** use communications technologies like **teleconferencing** (allowing two or more people at different locations to have a simultaneous conference) or **videoconferencing** (participants can also see and hear each other and share data).

- **Crowdsourcing** is a type of collaboration where an organization outsources a task to an undefined, generally large group of people in the form of an open call.
4. **E-learning and distance learning:** **E-learning** is learning supported by the Web, and **distance learning (DL)** refers to any learning situation where teachers and students do not meet face-to-face. E-learning provides tools for business users to enable their lifelong learning.
 5. **Virtual universities:** Online universities in which students take classes via the Internet at home or an off-site location. Virtual universities make it possible for students to obtain degrees while working full time, thus increasing their value to their firms.
 6. **Telecommuting:** The process whereby knowledge workers are able to work anywhere and anytime. Telecommuting provides flexibility for employees, with many benefits and some drawbacks.

Explain the importance of Information systems to Society.

The effects of information technology (IT) are wide-ranging, global, and disruptive. The IT profoundly impacts individuals, organizations, and societies, the global economy, and our physical environment. IT is making our world smaller, enabling more and more people to communicate, collaborate, and compete, thereby leveling the digital playing field.

The increased use of information technology has both positive and negative societal effects:

Positive Societal Effects:

- IT can **provide opportunities for people with disabilities**.
- IT can provide people with **flexibility in their work** (e.g., work from anywhere, anytime).
- **Robots will take over mundane chores**.
- IT will enable **improvements in healthcare**.

Negative Societal Effects:

- IT can **cause health problems for individuals**.
- IT can **place employees on constant call**.
- IT can potentially **misinform patients** about their health problems.

The growth of IT also contributes to the **digital divide**, which is the gap between those who have access to information and communications technology and those who do not.

Develop the plan for delivery application in M-commerce using social computing.

A specific plan for delivering an M-commerce application using social computing is not available in the provided sources. However, the sources define the necessary concepts and provide examples of how these technologies are currently used, which outlines the foundation and components required for such a plan.

M-Commerce and Social Computing Definition:

- **Mobile commerce (m-commerce)** refers to electronic commerce (EC) transactions that are conducted in a wireless environment, especially via the Internet.
- **Social computing** is a type of information technology that combines social behavior and information systems to create value. **Social commerce** is the delivery of electronic commerce activities and transactions through social computing.

M-Commerce Applications (Delivery Focus):

1. **Location-Based Commerce (L-Commerce):** M-commerce transactions targeted to individuals in specific locations, at specific times. A delivery application could use this to:
 - Request the nearest business or service (e.g., nearest available delivery driver).

- Provide **location-based advertising**, integrating the current locations and preferences of mobile users to send user-specific advertising messages (e.g., a flash deal for coffee when a user is near a café).
 - Utilize GPS (Global Positioning System) for navigation and tracking.
2. **Telemetry:** Wireless transmission and receipt of data gathered from remote sensors. A delivery application could use telemetry to monitor the location and status (e.g., engine temperature, speed) of delivery vehicles in real time.
 3. **Mobile Computing and Mobile Wallets:** Delivery requires quick and secure payment. **Mobile wallets (m-wallets)** are technology that allows users to make purchases with a single click from their mobile devices. This facilitates wireless payment systems for delivery agents.
 4. **Intrabusiness Applications:** These applications streamline workflow within the organization. In a delivery service, Mobile CRM systems and Sales Force Automation (SFA) would provide new opportunities to manage communications between the organization, the driver, and the customer.

Social Computing Integration (The Social Layer):

1. **Social Shopping/F-Commerce:** Leveraging social networks to facilitate transactions.
 - **Commerce-on Facebook (F-commerce)** allows transactions to occur completely inside Facebook, where vendors create Facebook stores for customers. A delivery service could allow customers to book and pay for delivery services without leaving the social platform.
 - **Group Shopping** sites like Groupon and LivingSocial, which offer deals based on collective purchasing. A delivery service could offer "group deals" for neighborhood delivery days.
2. **Social Intelligence/Conversational Marketing:** Companies must closely monitor social computing for feedback. A delivery application plan would integrate systems to monitor, collect, and analyze socially generated data (**social intelligence**) to conduct better marketing campaigns and improve service offerings. This supports **conversational marketing**, enabling customers to supply feedback via blogs, wikis, and social networking sites.

3. **Collaborative Consumption:** Social systems are used to facilitate peer-to-peer sharing or renting. A delivery app could function as a sharing marketplace, matching customers who need items moved with local drivers who have capacity, thus extracting value out of underutilized resources.

Explain the major security threats to information security and discuss the measures for controlling the same.

Information security is the protection of an organization's information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. There are two categories of threats: unintentional and deliberate.

The ten common types of **deliberate threats** to information systems are:

1. **Espionage or trespass:** Occurs when an unauthorized individual attempts to gain illegal access to organizational information.
2. **Information extortion:** Occurs when an attacker either threatens to steal, or actually steals, information from a company and demands payment for not stealing the information, for returning stolen information, or for agreeing not to disclose the information.
3. **Sabotage or vandalism:** Deliberate acts that involve defacing an organization's website or destroying systems.
4. **Theft of equipment or information:** Highly sensitive, classified, or proprietary data may be stolen from the organization.
5. **Identity theft:** Crime in which someone uses the personal information of others to create a false identity and then uses it for some fraud.
6. **Compromises to intellectual property:** Protecting intellectual property (property created by individuals or corporations protected under trade secret, patent, and copyright laws) is vital.
7. **Software attacks:** Malicious software (**malware**), such as viruses, worms, and Trojan horses.

8. **Alien software:** Clandestine software installed on a computer through duplicitous methods. This includes **spyware** (records keystrokes/passwords) and **spamware** (uses your computer as a launch platform for spammers).
9. **Supervisory control and data acquisition (SCADA) attacks:** SCADA systems control industrial equipment (like power grids, gas pipelines). Attacks on these systems pose serious threats to civilian populations.
10. **Cyberterrorism and cyberwarfare:** Cyberterrorism is a premeditated, politically motivated attack that results in violence against noncombatant targets. Cyberwarfare involves attacks on a country's information systems.

Measures for Controlling Security Threats

Organizations use controls (also called **countermeasures**) to protect information resources, categorized into three major types:

1. Physical Controls:

- **Purpose:** Prevent unauthorized individuals from gaining access to a company's facilities.
- **Examples:** Walls, doors, fencing, gates, locks, badges, guards, and alarm systems.

2. Access Controls:

- **Purpose:** Restrict unauthorized individuals from using information resources and are concerned with user identification.
- **Key Processes:**
 - **Authentication:** Determines the identity of the person requiring access. Methods include passwords, smart ID cards, and **biometrics** (measuring physiologic or behavioral characteristics, like fingerprints or voice patterns).
 - **Authorization:** Determines which actions, rights, or privileges the person has, based on verified identity. The principle of **least privilege** states that users be granted privilege for some activity only if there is a justifiable need.

3. Communications Controls (Network Controls):

- **Purpose:** Secure the movement of data across networks.
- **Examples:**
 - **Firewalls:** Systems (hardware, software, or both) that prevent a specific type of information from moving between untrusted networks (like the Internet) and private networks. They can be installed at the corporate gateway or on individual computers.
 - **Anti-malware systems (Antivirus software):** Software packages that identify and eliminate viruses, worms, and other malicious software.
 - **Whitelisting and Blacklisting:** **Blacklisting** identifies certain types of software that are *not allowed* to run. **Whitelisting** identifies acceptable software and permits *only* it to run.
 - **Encryption:** The process of converting an original message into a form that cannot be read by anyone except the intended receiver. The **Secure Socket Layer (SSL)** is an encryption standard used for secure transactions.
 - **Virtual Private Networks (VPNs):** Private networks that use a public network (usually the Internet) to securely connect users using encryption, often through a process called **tunneling**.

Explain CRM. Describe the different types of CRM with example.

Customer Relationship Management (CRM) is a customer-focused and customer-driven organizational strategy that concentrates on addressing customers' requirements for products and services and then providing high-quality, responsive service. Organizations focus on maximizing the number of high-value repeat customers while minimizing **customer churn** (loss of customers).

A successful CRM strategy is built on recognizing **customer touch points** (any interaction between a customer and an organization) and consolidating data about each customer. The objective is to achieve a **360-degree view of that customer**,

a complete data set on each customer made available to every unit of the business.

CRM systems generally lie along a continuum and are primarily divided into operational and analytical components:

1. Operational CRM Systems

These systems support the **front-office business processes**—sales, marketing, and service—that directly interact with customers.

Type of Application	Description	Example/Function	Source
Customer-Facing Applications	Areas where customers directly interact with the organization.	Sales Force Automation (SFA): Automatically records all aspects of a sales transaction process. Customer Service and Support: Uses methods like customer interaction centers (CICs) to manage customer communications.	
Customer-Touching Applications (e-CRM)	Applications and technologies with which customers interact and typically help themselves.	Personalized Web Pages: Optimized pages based on past behavior. Campaign Management: Helps organizations plan campaigns that send the right messages to the right people. Loyalty Programs: Reward customers to influence future behavior.	

2. Analytical CRM Systems

These systems analyze customer behavior and perceptions to provide **actionable business intelligence**. Analytical systems use **data warehouses**, **data mining**, **decision support**, and other business intelligence technologies.

- **Example Applications:** Designing and executing **targeted marketing campaigns**; predicting customer acquisition, **cross selling** (marketing additional products based on a previous purchase), and **up selling** (marketing

higher value products); providing input into decisions relating to pricing and product development; and providing financial forecasting and customer profitability analysis.

3. Collaborative CRM Systems

These provide effective and efficient interactive communication with the customer throughout the entire organization, integrating communications between the organization and its customers in all aspects of marketing, sales, and customer support.

- **Example:** Utilizing Web 2.0 applications such as blogs and Wikis to enable customers to provide direct feedback to the organization, particularly valuable for new product development.

4. Other Deployment Types

Type	Description	Example	Source
On-Demand CRM Systems	Hosted by an external vendor in the vendor's data center.	Salesforce.	
Mobile CRM Systems	Interactive CRM systems where communications related to sales, marketing, and customer service activities are conducted through a mobile medium.	Sales force automation (SFA) provides new opportunities in the field.	
Open-Source CRM Systems	CRM software whose source code is available to developers and users.	SugarCRM.	

What is Cloud Computing? Explain its models.

Cloud computing is a type of computing that delivers convenient, on-demand, pay-as-you-go access for multiple customers to a shared pool of configurable computing resources (e.g., servers, networks, storage, applications, and services) that can be rapidly and easily accessed over the Internet. The fundamental

concept is that tasks are performed by computers physically removed from the user and accessed over a network, in particular the Internet.

Cloud computing enables businesses to rapidly adjust their processes, products, and services to meet the changing needs of the market.

Essential Characteristics of Cloud Computing

Cloud computing generally includes the characteristics of **grid computing** (applying unused processing resources of geographically dispersed computers to form a virtual supercomputer) and **utility computing** (a pay-as-you-go service).

The essential characteristics of cloud computing include:

- **On-demand self-service:** Users can provision computing capabilities, such as server time and network storage, without human interaction.
- **Broad network access:** Services are available over the network.
- **Resource pooling:** Computing resources are pooled to serve multiple users, with resources dynamically assigned or reassigned according to customer demand.
- **Rapid elasticity and measured service:** Resources can be rapidly and elastically provisioned and released to scale outward and inward rapidly as needed. Customers pay only for the resources actually consumed.
- **Virtualization:** Cloud computing typically occurs on virtualized servers. **Virtualization** technology uses software to create virtual versions of computing resources.

Cloud Deployment Models (Types of Clouds)

There are four deployment models describing how the cloud is located and managed:

1. **Public Clouds:** Shared, easily accessible, multi-customer IT infrastructures available non-exclusively to any entity (individuals, groups, and/or organizations).
2. **Private Clouds (Internal/Corporate Clouds):** IT infrastructures that are accessible only by a single entity, or by an exclusive group of related entities

that share the same purpose and requirements (e.g., all business units within a single organization).

3. **Hybrid Clouds:** Composed of public and private clouds that remain unique entities but are bound together, offering the benefits of multiple deployment models.
4. **Vertical Clouds:** Cloud infrastructure designed for and focused on serving specific industries.

Cloud Service Models (Types of Services)

Cloud computing providers offer three main service models (sometimes called the "stack"):

1. **Infrastructure-as-a-Service (IaaS) Model:** Cloud computing providers offer remotely accessible servers, networks, and storage capacity. Customers rent these computing resources and control the operating system and installed applications.
2. **Platform-as-a-Service (PaaS) Model:** Customers rent servers, operating systems, storage, a database, software development technologies (such as Java and .NET), and network capacity over the Internet. This allows customers to develop, run, and manage applications without the complexity of managing the infrastructure.
3. **Software-as-a-Service (SaaS) Delivery Model:** Cloud computing vendors provide software that is specific to their customers' requirements. Vendors host the applications and provide them as a service to customers over a network, typically the Internet. This model allows applications to be accessible from various client devices through a thin-client interface.

Write note on e-business.

Electronic business (e-business) is a broad concept that encompasses more than just the buying and selling of goods and services. It refers to servicing customers, collaborating with business partners, and performing electronic transactions within an organization.

E-commerce, a narrower concept than e-business, describes the process of buying, selling, transferring, or exchanging products, services, or information via computer networks, including the Internet.

Degree of Digitization

Electronic commerce can take several forms depending on the **degree of digitization** (the extent to which the commerce has been transformed from physical to digital, relating to the product/service sold and the delivery agent):

- **Brick-and-mortar organizations:** Organizations where the product, the process, and the delivery agent are all physical.
- **Virtual (or pure-play) organizations:** Companies engaged only in EC where all dimensions (product, process, delivery) are digital.
- **Clicks-and-mortar organizations (clicks-and-bricks):** Organizations that conduct some e-commerce activities, yet their primary business is carried out in the physical world.

Types of Electronic Commerce

There are six common types of e-commerce, conducted between and among various parties:

1. **Business-to-Consumer (B2C) EC:** Sellers are organizations, and buyers are individuals. (Example: Buying a book from Amazon.com).
2. **Business-to-Business (B2B) EC:** Both sellers and buyers are business organizations. B2B comprises the vast majority of EC volume.
3. **Consumer-to-Consumer (C2C) EC:** An individual sells products or services to other individuals. Major strategies are auctions (e.g., eBay) and classified ads (e.g., Craigslist).
4. **Business-to-Employee (B2E) EC:** An organization uses EC internally to provide information and services to its employees (e.g., managing benefits or taking training classes electronically).
5. **E-Government:** The use of Internet technology to deliver public services to citizens (**Government-to-Citizen or G2C**) and to business partners/suppliers (**Government-to-Business or G2B**).

6. **Mobile Commerce (M-commerce):** E-commerce that is conducted entirely in a wireless environment.

Major E-Commerce Mechanisms

Businesses and customers use several mechanisms to buy and sell on the Internet:

- **Electronic Catalogs:** Consist of a product database, a directory/search capabilities, and a presentation function; the backbone of most e-commerce sites.
- **Electronic Auctions (e-auctions):** A competitive buying and selling process where prices are determined dynamically by competitive bidding, using the Internet as an efficient infrastructure. Auctions can be **forward auctions** (seller to many potential buyers) or **reverse auctions** (one buyer seeks to buy from many suppliers).
- **Electronic Storefronts:** A Web site that represents a single store (e.g., Dell.com).
- **Electronic Malls (e-malls):** A collection of individual shops under one Internet address (e.g., a cybermall).
- **Electronic Marketplaces (e-marketplaces):** A central, virtual market space where many buyers and many sellers can conduct e-commerce and e-business activities; typically associated with B2B.

Explain the phases of the system development lifecycle with example.

The **systems development life cycle (SDLC)** is the traditional systems development method that organizations use for large-scale IT projects. It is a structured framework consisting of sequential processes by which information systems are developed. It consists of six processes, each with clearly defined tasks:

Phase	Primary Tasks	Deliverable/Focus	Source
1. Systems Investigation	Addresses the business problem or opportunity. The main task is the feasibility study .	Determine whether the project is achievable and worth undertaking.	
2. Systems Analysis	Examines the business problem the organization plans to solve. Main purpose is to gather information about the existing system to determine the requirements for the new system .	A set of system requirements.	
3. Systems Design	Describes <i>how</i> the new system will resolve the business problem. This typically addresses both physical and logical design.	The set of technical system specifications .	
4. Programming and Testing	Programming translates the design specifications into computer code. Testing checks if the code produces expected and desired results and detects errors, or bugs .	The new application.	
5. Implementation	The process of converting from the old system to the new system.	A properly working application.	
6. Operation and Maintenance	Involves auditing the system and performing three types of maintenance: debugging the program, updating the system, and adding new functions when needed.	Continued successful operation and adaptation.	

Feasibility Study (Part of Systems Investigation):

The investigation phase must determine the technical, economic, and behavioral feasibility of the project:

- **Technical feasibility:** Determines whether the company can develop and/or acquire the hardware, software, and communications components needed.
- **Economic feasibility:** Determines whether the project is an acceptable financial risk and whether the organization has the necessary time and money to successfully complete the project. This often involves methods like **Return on Investment (ROI)**, **Net Present Value (NPV)**, and **Breakeven analysis**.
- **Behavioral feasibility:** Addresses the human issues of the systems development project (e.g., acceptance by systems stakeholders).

Implementation Conversion Strategies (Part of Implementation):

During implementation, the conversion from the old system to the new system can follow three major strategies:

1. **Direct conversion:** The old system is cut off, and the new system is turned on at a certain date.
2. **Pilot conversion:** Introduces the new system in one part of the organization on a trial basis.
3. **Phased conversion:** Introduces components of the new system in stages, until the entire new system is operational.

Compare and contrast Web 1.0, 2.0, 3.0 with example.

Web 1.0 was the first generation of the World Wide Web, which first appeared in 1990.

- Key developments of Web 1.0 were the creation of Web sites and the commercialization of the Web.
- Users typically had **minimal interaction** with Web 1.0 sites.
- Users **passively receive information** from those sites.

Web 2.0 is a popular term for the loose collection of information technologies and applications, plus the Web sites that use them.

- Web 2.0 sites **enrich the user experience** by encouraging user participation, social interaction, and collaboration.
- Unlike Web 1.0, Web 2.0 sites facilitate information sharing, **user-centered design**, and collaboration.
- Web 2.0 sites often harness **collective intelligence** (e.g., wikis).
- They deliver functionality as services rather than packaged software (e.g., Web services).
- They feature **remixable applications and data** (e.g., mashups).
- **Examples of Web 2.0 tools** include AJAX (which reloads portions of pages with fresh data), **tagging** (keywords describing content, forming folksonomies), **blogs** (personal journals), **microblogging** (short messages, like Twitter), **wikis** (user-posted editable content), and **mashups** (mixing content from different sites to create new content).

Web 3.0 is not defined or described in the provided sources.

Compare between mobile applications like WhatsApp and Telegram on various parameters like multimedia, security, privacy, storage, data usage, notifications

The provided sources do not contain comparative information regarding the specific mobile applications WhatsApp or Telegram, nor do they detail the various technical parameters such as multimedia, security, privacy, storage, data usage, or notifications for these platforms.

Compare and contrast basic types of reports which are closely associated with

FAIS and ERP system.

Functional Area Information Systems (FAISs) support a particular functional area within the organization. **Enterprise Resource Planning (ERP) systems** are designed to correct the lack of communication among FAISs by tightly integrating these systems via a common database, supporting the entire organization. All information systems, including FAIS and ERP systems, produce reports. These reports generally fall into three categories:

Report Type	Description and Characteristics	Example
Routine Reports	Produced at scheduled intervals. They are useful to an organization but may lack flexibility concerning timing or inclusion of special information.	Hourly quality control reports or daily reports on absenteeism rates.
Ad-Hoc (On-Demand) Reports	Out-of-the routine reports that often contain special information not included in routine reports. Managers require these reports when they need specific information or routine information at different times.	A chief financial officer might want to monitor cash flow and cash on hand. Drill-down reports are a type of ad-hoc report that show a greater level of detail than is included in routine reports. Key-indicator reports are another type that summarize the performance of critical activities.
Exception Reports	Include only information that falls outside certain predetermined threshold standards. They are valuable for detecting issues like fraudulent transactions or poor performance.	A report identifying only those cases where sales fell outside an established threshold, such as more than 20 percent short of the quota. Comparative reports compare performances of different business units or times.

What is Big Data? What are the various challenges and characteristics of Big Data?

Definition of Big Data

Big Data refers to the superabundance of data available today. The technology research firm Gartner defines Big Data as **diverse, high-volume, high-velocity information assets** that require new forms of processing to enable enhanced decision making, insight discovery, and process optimization. The Big Data Institute defines Big Data as vast data sets that:

- Exhibit **variety**.
- Include **structured, unstructured, and semi-structured** data.
- Are generated at **high velocity** with an uncertain pattern.
- Do not fit neatly into traditional, structured, relational databases.
- Can be captured, processed, transformed, and analyzed in a reasonable amount of time only by sophisticated information systems.

Big Data generally consists of traditional enterprise data, machine-generated/sensor data, social data, and images captured by billions of devices.

Characteristics of Big Data

Big Data has three distinct characteristics that distinguish it from traditional data: **Volume, Velocity, and Variety**.

1. **Volume:** Big Data consists of vast quantities of data. The world generates exabytes of data each year (an exabyte is one trillion terabytes), and the amount of data produced worldwide is increasing by 50 percent each year. The sheer volume makes Big Data incredibly valuable.
2. **Velocity:** Big Data flows into an organization at incredible speeds. High velocity enables online retailers to compile histories on customer clicks and interactions in real time, giving them a competitive advantage if they can utilize that information quickly.
3. **Variety:** Big Data includes a huge variety of different data in differing formats. Unlike traditional data formats, which are structured and change slowly, Big Data formats change rapidly, including satellite imagery, broadcast audio streams, Web page content, and comments posted on social networks.

Challenges and Management of Big Data

Managing Big Data is a very real problem every business faces. Challenges arise because:

- The amount of data increases exponentially with time, forcing companies to develop a strategy for managing these data effectively.
- The data are often unstructured.
- Traditional, structured, relational databases cannot neatly contain Big Data.

To address these challenges, organizations are leveraging Big Data in several ways:

- Using **NoSQL databases** (which manipulate structured and unstructured data, and inconsistent data) as an alternative to traditional relational databases.
- **Creating Transparency** by making Big Data easier for relevant stakeholders to access quickly.
- **Enabling Experimentation** by collecting accurate performance data to set up controlled experiments.
- **Replacing/Supporting Human Decision Making with Automated Algorithms** using sophisticated analytics to improve decisions and minimize risks.
- **Innovating New Business Models, Products, and Services** by using data to create new offerings.
- **Analyzing Far More Data** (sometimes entire datasets) rather than relying only on sampling.

Describe various Cloud Computing Models and highlight their evolution

Evolution of IT Infrastructure

Cloud computing represents the latest stage in the evolution of modern IT infrastructure, which began in the 1950s:

1. **Stand-alone mainframes:** Used initially in engineering and accounting departments, housed securely, with access limited to MIS personnel.

2. **Mainframe and dumb terminals:** Dumb terminals were placed in user departments, enabling users to input programs into the mainframe from their location (remote job entry).
3. **Stand-alone personal computers:** Appeared in the late 1970s/early 1980s, brought by users to improve personal productivity, eventually leading organizations to support them.
4. **Local area networks (client/server computing):** Personal computers were connected into LANs, then connected to the mainframe.
5. **Enterprise computing:** Organizations used networking standards (like TCP/IP) to integrate different kinds of networks throughout the firm, allowing software and data to flow seamlessly across the enterprise.
6. **Cloud computing and mobile computing:** Providing convenient, on-demand, pay-as-you-go access to a shared pool of computing resources (servers, storage, applications, and services) over a network, typically the Internet.

Cloud Service Models (The Stack)

Cloud computing providers offer three main service models:

1. Infrastructure-as-a-Service (IaaS) Model:

- **Description:** Cloud providers offer remotely accessible servers, networks, and storage capacity. Customers rent these resources.
- **Customer Responsibility:** Customers control the operating system and installed applications.

2. Platform-as-a-Service (PaaS) Model:

- **Description:** Customers rent servers, operating systems, storage, a database, software development technologies (such as Java and .NET), and network capacity over the Internet.
- **Utility:** This model allows customers to develop, run, and manage applications without the complexity of managing the infrastructure.

3. Software-as-a-Service (SaaS) Delivery Model:

- **Description:** Vendors provide software specific to their customers' requirements. The applications are hosted by vendors and delivered as a

service over a network, typically the Internet.

- **Utility:** SaaS allows applications to be accessible from various client devices through a thin-client interface. A key differentiator is its ability to scale, adjusting server usage to meet demand transparently to the user.

Cloud Deployment Models (Types of Clouds)

There are four deployment models describing how the cloud is located and managed:

1. **Public Clouds:** Shared, easily accessible, multi-customer IT infrastructures available non-exclusively to any entity (individuals, groups, and/or organizations).
2. **Private Clouds (Internal/Corporate Clouds):** IT infrastructures that are accessible only by a single entity, or by an exclusive group of related entities that share the same purpose and requirements (e.g., all business units within a single organization).
3. **Hybrid Clouds:** Composed of public and private clouds that remain unique entities but are bound together, offering the benefits of multiple deployment models.
4. **Vertical Clouds:** Cloud infrastructure designed for and focused on serving specific industries (e.g., construction, finance, or insurance).

Illustrate Knowledge Management life cycle

Knowledge Management (KM) is a process that helps organizations identify, select, organize, disseminate, transfer, and apply information and expertise that are part of the organization's memory and that typically reside in an unstructured manner. **Knowledge management systems (KMSs)** use information technologies (Internet, intranets, databases) to systematize, enhance, and expedite this process.

A functioning KMS follows a cycle that consists of six steps:

1. **Create knowledge:** This is the phase where new knowledge is generated.

2. **Capture knowledge:** The new knowledge must be identified as valuable and be represented in a reasonable way.
3. **Refine knowledge:** New knowledge must be placed in context so that it is actionable, which involves capturing tacit qualities (human insights) along with explicit facts.
4. **Store knowledge:** Useful knowledge must then be stored in a reasonable format in a knowledge repository so that others in the organization can access it.
5. **Manage knowledge:** The knowledge must be continuously maintained and kept current, and reviewed regularly to verify that it is relevant and accurate.
6. **Disseminate knowledge:** Knowledge must be made available in a useful format to anyone in the organization who needs it, anywhere and anytime.

Illustrate the different types of knowledge and explain four modes of knowledge generation of Big Data.

Types of Knowledge

Knowledge is distinct from data (collection of facts) and information (organized data); knowledge is information that is **contextual, relevant, and useful**.

Intellectual capital (or intellectual assets) is another term for knowledge.

1. **Explicit knowledge:** Deals with more objective, rational, and technical knowledge. This knowledge has been **codified (documented)** in a form that can be distributed to others or transformed into a process or a strategy. Examples include organizational policies, procedural guides, reports, strategies, and IT infrastructure.
2. **Tacit knowledge:** The cumulative store of subjective or **experiential learning**. It is highly personal, hard to formalize, and generally undocumented.

Four Modes of Leveraging Big Data

While the sources do not explicitly define "four modes of knowledge generation of Big Data," they do outline **six broadly applicable ways to leverage Big Data to gain value**. Four of these ways are:

1. **Creating Transparency:** Making Big Data easier for relevant stakeholders to access in a timely manner creates tremendous business value. In manufacturing, integrating data from R&D, engineering, and manufacturing enables concurrent engineering, reducing time to market and improving quality.
2. **Enabling Experimentation:** Organizations can collect more accurate and detailed performance data (in real or near-real time) on everything from inventories to sick days. IT enables organizations to set up controlled experiments, such as Amazon's A/B experiments on its website design.
3. **Replacing/Supporting Human Decision Making with Automated Algorithms:** Sophisticated analytics can substantially improve decision making, minimize risks, and unearth valuable insights. For example, tax agencies use automated risk-analysis software tools to identify tax returns that require further examination.
4. **Innovating New Business Models, Products, and Services:** Big Data enables companies to create new products and services, enhance existing ones, and invent entirely new business models. The emergence of real-time location data, for instance, created an entirely new set of location-based services ranging from navigation to pricing insurance based on driving habits.

Explain in detail computer-based information systems.

A **computer-based information system (CBIS)** is an information system that uses computer technology to perform some or all of its intended tasks. Because most modern information systems are computerized, the term "information system" is typically used synonymously with CBIS.

Components of CBIS

The basic components of CBIS are called the **information technology components**:

- **Hardware** (e.g., processor, monitor).
- **Software** (program or collection of programs that enable hardware to process data).
- **Database** (collection of related files or tables containing data).
- **Networks** (connecting systems that permit different computers to share resources).

These components form the **information technology platform**. IT personnel use these components to perform IT services—developing systems, managing security and risk, and managing data. The IT components plus IT services comprise the organization's **information technology infrastructure**.

Capabilities of CBIS

CBISs have several important capabilities:

- Perform high-speed, high-volume numerical computation.
- Provide fast, accurate communication and collaboration within and among organizations.
- Store huge amounts of information in an accessible, yet small, space.
- Allow quick and inexpensive access to vast amounts of information worldwide.
- Increase the effectiveness and efficiency of people working in large groups in one location or in several locations.
- Automate both semi-automatic business processes and manual tasks.
- Facilitate the interpretation of vast amounts of data.
- Apply advanced analytical techniques to transform data into information and knowledge.

Types of CBIS in Organizations

CBIS can support different organizational scopes, illustrated hierarchically:

1. Systems Supporting the Entire Organization:

- **Transaction Processing Systems (TPSs):** Support the monitoring, collection, storage, and processing of data from the organization's basic business transactions. These systems provide the inputs into the organization's database.
- **Enterprise Resource Planning (ERP) Systems:** Correct a lack of communication among functional area systems by tightly integrating them via a common database.

2. Systems Supporting Functional Areas:

- **Functional Area Information Systems (FAISs):** Support a particular functional area within the organization (e.g., accounting IS, finance IS, marketing IS, human resources IS, production/operations management IS).

3. Interorganizational Systems (IOSs):

- **Interorganizational Information Systems (IOSs):** Connect two or more organizations. The best-known examples are:
 - **Supply Chain Management (SCM) Systems:** Support the flow of materials, information, money, and services from suppliers through the organization to the end customers.
 - **Electronic Commerce (E-commerce) Systems:** Enable organizations to conduct transactions with other organizations (B2B) and with customers (B2C).

4. Business Intelligence and Decision Support Systems:

- **Business Intelligence (BI) Systems:** Provide computer-based support for complex, nonroutine decisions, primarily for middle managers and knowledge workers.
- **Expert Systems (ESs):** Attempt to duplicate the work of human experts by applying reasoning capabilities, knowledge, and expertise within a specific domain.
- **Dashboards:** A special form of IS that supports all managers by providing rapid access to timely information and direct access to structured information in the form of reports.

What are the different types of MIS?

The term **Management Information Systems (MIS)** is often used broadly, sometimes interchangeably with the Information Systems (IS) Department or the Information Technology Department. The MIS functional area deals with the planning for—and the development, management, and use of—information technology tools to help people perform all tasks related to information processing and management.

Modern organizations employ many different types of Computer-Based Information Systems (CBISs), which serve various organizational functions and hierarchies. These types, which collectively form the operational and decision-making infrastructure often associated with MIS, include:

1. **Transaction Processing Systems (TPSs):** These systems support the monitoring, collection, storage, and processing of data from the organization's basic business transactions. TPSs are the most fundamental organizational IS and provide the inputs (data) into the organization's database.
2. **Functional Area Information Systems (FAISs):** These departmental systems support a particular functional area within the organization by increasing each area's internal efficiency and effectiveness. Examples include:
 - Accounting IS
 - Finance IS
 - Marketing IS
 - Human Resources IS
 - Production/Operations Management (POM) IS.
3. **Enterprise Resource Planning (ERP) Systems:** These systems correct a lack of communication among the FAISs (which often exist as information silos) by tightly integrating the functional area ISs via a common database.
4. **Business Intelligence (BI) Systems:** These systems provide computer-based support for complex, nonroutine decisions, primarily for middle managers and

knowledge workers. They enable users to perform their own data analysis and are often used with a data warehouse.

5. **Dashboards (Digital Dashboards):** These are a special form of IS that support **all managers** of the organization by providing rapid access to timely information and direct access to structured information in the form of reports.
6. **Expert Systems (ESs):** These attempt to duplicate the work of human experts by applying reasoning capabilities, knowledge, and expertise within a specific domain, primarily involving decision making.
7. **Interorganizational Information Systems (IOSs):** These systems connect two or more organizations, such as:
 - **Supply Chain Management (SCM) Systems:** Manage the flow of materials, information, money, and services from raw material suppliers through factories and warehouses to the end customers.
 - **Electronic Commerce (E-commerce) Systems:** Enable organizations to conduct transactions with other organizations (B2B) and with customers (B2C).

How is data governance achieved in case of MIS?

The MIS function, which manages the organization's data and databases, must ensure data governance. **Data governance** is an approach to managing information across an entire organization. It involves a formal set of business processes and policies designed to ensure that data are handled in a certain, well-defined fashion, following unambiguous rules for creating, collecting, handling, and protecting information.

The objective of data governance is to make information ** available, transparent, and useful** for authorized people, from the moment it enters an organization until it is outdated and deleted.

Master Data Management (MDM) as a Strategy for Data Governance:

One primary strategy for implementing data governance is **master data management**.

- MDM is a process that spans all organizational business processes and applications.
- It provides companies with the ability to store, maintain, exchange, and synchronize a consistent, accurate, and timely **“single version of the truth”** for the company’s core **master data**.
- **Master data** are core data (like customer, product, employee, or location data) that apply to multiple transactions and are used to categorize, aggregate, and evaluate **transaction data**.

The Need for Data Governance (The Data Management Problem):

Data governance is necessary because managing data is inherently difficult due to several challenges that impact the MIS function:

- **Data Scarcity and Location:** Data are scattered throughout organizations, collected by many individuals, and stored in numerous servers, locations, computing systems, databases, and formats.
- **Multiple Sources:** Data are generated from various sources, including internal sources (corporate databases), personal sources (opinions), and external sources (Web sites, government reports).
- **Inconsistency and Repetition:** Information systems that support specific business processes (like transaction processing or SCM) impose unique requirements on data, resulting in repetition and conflicts across the organization.
- **Regulatory Compliance:** Federal regulations, such as Sarbanes-Oxley, mandate that public companies better account for how they manage information and hold CEOs/CFOs personally responsible for disclosures, forcing a focus on data management policies.

Analyse briefly between Web 2.0 and Web 3.0? to highlight the difference

Feature	Web 2.0	Web 3.0	Source
Description	A popular term for the loose collection of information	Not defined or described in the	

Feature	Web 2.0	Web 3.0	Source
	technologies, applications, and Web sites that enrich the user experience by encouraging user participation, social interaction, and collaboration.	provided sources.	
Key Focus	User participation, user-centered design, collaboration, information sharing, and harnessing collective intelligence .		
Data/Functionality	Delivers functionality as services (Web services) and features remixable applications and data (mashups).		
Technology Examples	AJAX, tagging, blogs, microblogging, wikis, social networking sites, and mashups.		

Note: The provided sources define and discuss **Web 2.0** but do not offer a definition, description, or analysis of **Web 3.0**.

Evaluate the MIS Hierarchy to comment on Decision Support System.

The MIS Hierarchy (or different types of Computer-Based Information Systems) demonstrates how organizational systems support various levels of management and decision-making complexity. The **Decision Support System (DSS)** primarily targets **middle managers and knowledge workers** dealing with **semi-structured and unstructured problems**.

MIS Hierarchy and DSS Placement:

Systems generally progress from handling routine transactions (low level) to supporting complex decisions (high level):

- **Transaction Processing Systems (TPSs):** Handle routine transactions and provide data inputs.
- **Functional Area Information Systems (FAISs):** Support specific departmental operations.
- **Decision Support Systems (DSSs) / Business Intelligence (BI) Systems:** Provide computer-based support for complex, nonroutine, semi-structured, and unstructured decisions.
- **Dashboards (Executive Dashboards):** Support all managers and executives by providing rapid access to timely, summarized information.

Commentary on Decision Support Systems (DSSs):

1. **Purpose and Structure:** DSSs combine **models and data** in an attempt to analyze **semi-structured problems** and some **unstructured problems** that involve extensive user involvement. DSSs are crucial because decision-making is becoming more difficult due to the constantly increasing number of alternatives, time pressure, and increased uncertainty/complexity.
2. **Functionality:** DSSs enable business managers and analysts to access data interactively, manipulate these data, and conduct appropriate analyses. They employ mathematical models and offer three related analytical capabilities:
 - **Sensitivity Analysis:** Studies the impact that changes in one or more parts of a model have on other parts.
 - **What-if Analysis:** Predicts the impact of changes in assumptions (input data) on the proposed solution.
 - **Goal-Seeking Analysis:** Calculates the values of the inputs necessary to achieve a desired goal (not explicitly mentioned in the provided text as part of DSS, but listed as a related capability).
3. **Hierarchy Context:** DSSs are specifically designed for decisions that are **not routine or repetitive** (structured decisions, usually handled by TPS or FAIS). They fall under **Management Control** and **Strategic Planning** on the decision support framework, focusing on how to construct models and interpret data to make complex choices.

List the main difference between Wireless and Wired Technologies?

The sources refer to wired and wireless technologies primarily through the lens of network and communication media.

Feature	Wired Technologies (Wireline Media/Cable Media)	Wireless Technologies (Broadcast Media/Wireless Media)	Source
Definition	Communications channels that use physical wires or cables to transmit data and information.	Communications channels that use electromagnetic media (the “airwaves”) to transmit data.	
Medium	Fiber-optic cable, coaxial cable, twisted-pair wire.	Radio, microwave, satellite, infrared.	
Mobility	The term wireless means exactly what it says: without wires . Some wireless networks are also mobile (changing location over time), while others are fixed.	Mobile computing refers to a real-time, wireless connection between a mobile device and other computing environments.	
Security	Generally considered more secure as they rely on physical lines.	Inherently a nonsecure broadcast communications medium ; transmissions can be intercepted by anyone close enough with appropriate equipment.	

Give an understanding on types of Control to achieve Security.

Controls (also called **countermeasures**) are defense mechanisms organizations use to protect their information resources and achieve security. The goal of security is to protect against criminal activity, danger, damage, and/or loss.

Controls are categorized into three major types:

1. Physical Controls:

- **Purpose:** Prevent unauthorized individuals from gaining access to a company's facilities.
- **Examples:** Walls, doors, fencing, gates, locks, badges, guards, and alarm systems. More sophisticated examples include pressure sensors, temperature sensors, and motion detectors.

2. Access Controls:

- **Purpose:** Restrict unauthorized individuals from using information resources and focus on user identification.
- **Key Processes:**
 - **Authentication:** Determines the identity of the person requiring access. Methods include passwords, smart ID cards, tokens, and biometrics (measuring physiologic or behavioral characteristics).
 - **Authorization:** Determines which actions, rights, or privileges the authenticated person has. The principle of **least privilege** grants users only the necessary privilege for a justified need.

3. Communications Controls (Network Controls):

- **Purpose:** Secure the movement of data across networks.
- **Examples:**
 - **Firewalls:** Systems (hardware, software, or both) that prevent specific information types from moving between untrusted networks (like the Internet) and private networks.
 - **Anti-malware systems (Antivirus software):** Identify and eliminate viruses, worms, and malicious software.
 - **Whitelisting and Blacklisting:** Whitelisting identifies acceptable software and permits only it to run; Blacklisting identifies software that is not allowed to run.
 - **Encryption:** Converting a message into a form readable only by the intended receiver. **Secure Socket Layer (SSL)** is an encryption

standard used for secure transactions.

- **Virtual Private Networks (VPNs):** Private networks that use a public network (the Internet) securely via encryption and **tunneling**.

What is Mobile Commerce? What are the new challenges that it has introduced in business?

Mobile Commerce (M-commerce) Definition: Mobile commerce (m-commerce) refers to electronic commerce (EC) transactions that are conducted in a **wireless environment**, especially via the Internet. M-commerce creates opportunities for businesses to deliver new services to existing customers and attract new customers.

New Challenges Introduced in Business (Operational and Security):

1. **Security and Vulnerability:** Wireless is an **inherently nonsecure broadcast communications medium**. Transmissions can be intercepted by anyone nearby with the appropriate equipment. Major threats include:
 - **Security of Mobile Devices:** Smartphones can be used to copy and pass on confidential information (e.g., snapping pictures of secret technology).
 - **Physical and Logical Interception:** Threats involve **rogue access points** (used by attackers to set up unauthorized access points), **war driving** (locating WLANs while driving), and **radio-frequency (RF) jamming** (interfering with wireless network transmissions).
2. **Channel Conflict in Retail:** Mobile technology enables **showrooming**, where customers examine a product in a physical store but buy it online cheaper. This threatens traditional physical retailers.
3. **Management of New Payment Systems:** There is a "huge battle underway" to provide a mobile, digital wallet to replace physical wallets, involving conflicts over who will collect lucrative transaction fees from retailers.
4. **Privacy Concerns with Location-Based Services (L-commerce):** M-commerce applications often rely on location-based commerce (L-

commerce), where transactions are targeted to individuals in specific locations. This raises **privacy issues** because it means users' locations are being tracked.

What do you mean by CRM? Give its types and relate the role of SC on CRM.

Customer Relationship Management (CRM) Definition: Customer relationship management (CRM) is a customer-focused and customer-driven **organizational strategy**. Organizations concentrate on assessing customers' requirements for products and services and then providing high-quality, responsive service. The goal of CRM is to maximize the number of high-value repeat customers while minimizing **customer churn** (loss of customers). CRM systems aim to achieve **customer intimacy** by marketing to each customer individually.

Types of CRM Systems:

CRM systems generally fall into two major components, plus other deployment types:

1. Operational CRM Systems:

- **Focus:** Support **front-office business processes** (sales, marketing, and service) that directly interact with customers.
- **Components:** **Customer-facing applications** (e.g., Sales Force Automation, Customer Service and Support) and **Customer-touching applications** (e.g., Personalized Web Pages, FAQs, Loyalty Programs).

2. Analytical CRM Systems:

- **Focus:** Provide **business intelligence** by analyzing customer behavior and perceptions.
- **Utility:** Used for designing targeted marketing campaigns, increasing customer acquisition/cross selling/up selling, and performing financial forecasting and customer profitability analysis. They leverage tools like data warehouses and data mining.

3. Other Types:

- **Collaborative CRM Systems:** Provide effective interactive communication with the customer throughout the entire organization by integrating communication between the organization and customers in all aspects of marketing, sales, and customer support.
- **On-Demand CRM Systems:** Hosted by an external vendor (SaaS model).
- **Mobile CRM Systems:** Interactive systems conducted through a mobile medium.
- **Open-Source CRM Systems:** CRM software whose source code is available at no cost.

Relating the Role of Supply Chain (SC) on CRM: Supply Chain Management (SCM) systems and **CRM** systems are both critical to organizational success as they manage external relationships (suppliers and customers, respectively).

- **SCM** manages the upstream, internal, and downstream flows of materials, information, money, and services from suppliers to the end customers.
- **CRM** focuses specifically on managing the relationship with the **downstream end of the supply chain: the customers**.

The role of SCM data and systems often supports CRM objectives indirectly:

1. **Enhanced Service and Inventory Velocity:** Efficient SCM, supported by IT, increases **inventory velocity** (the speed products can be delivered). Fast and accurate delivery, supported by SCM visibility, is crucial for good customer service and retention, which are primary CRM goals.
2. **Financial Reporting and Controls:** CRM systems can track documents from a sales opportunity to an invoice, providing financial and accounting managers the data to monitor the flow. SCM also ensures that the necessary financial data related to inventory and cost of goods sold (received from supply chain partners) are available for reporting.
3. **Marketing Insights:** Marketing managers gain valuable sales information from supply chain partners through supporting information systems. Understanding how downstream SCM activities relate to previous operations is critical for marketing.

Evaluate the role of Confidentiality. Integrity and Availability in order to achieve security.

While the sources do not explicitly use the term "CIA triad" (Confidentiality, Integrity, and Availability), they define information security as the protection of information and systems from **unauthorized access, use, disclosure, disruption, modification, or destruction**. These terms directly correspond to the principles of Confidentiality, Integrity, and Availability:

Security Goal	Description (based on source definitions)	Role in Achieving Security	Source
Confidentiality (Preventing Unauthorized Access/Disclosure)	Protecting information from unauthorized access . This includes protecting data from disclosure or espionage .	Achieved primarily through Access Controls (Authentication, Authorization) and Communication Controls like Encryption and VPNs . Protecting customer data is critical for marketing and finance.	
Integrity (Preventing Unauthorized Modification/Destruction)	Protecting information from unauthorized modification or destruction .	Achieved by using controls to ensure the accuracy and reliability of data. Auditing information systems examines data integrity, security, and proper functioning. This is essential in a TPS to record results accurately .	
Availability (Preventing Disruption)	Ensuring that systems are protected from disruption .	Achieved through Business Continuity Planning (BCP) and the provision of hot sites	

Security Goal	Description (based on source definitions)	Role in Achieving Security	Source
		or warm sites to ensure IT operations can quickly recover after a disaster. ERP systems are designed to improve efficiency, which relies on high system availability.	

The overall role of these three concepts is to ensure the complete protection of an organization's information assets against myriad threats, which include deliberate attacks like cyberterrorism, theft, and software attacks.

Highlight how company can use Big data to gain competitive advantage?

Big Data refers to the diverse, high-volume, high-velocity information assets that require new processing forms to enable enhanced decision making, insight discovery, and process optimization. By effectively managing and leveraging Big Data, organizations can gain a significant competitive advantage.

Organizations can leverage Big Data in six broadly applicable ways to gain value and competitive advantage:

1. **Creating Transparency (Access and Insight):** Simply making Big Data easier for relevant stakeholders to access quickly creates tremendous business value. In manufacturing, integrating data across R&D, engineering, and manufacturing enables **concurrent engineering**, significantly reducing time to market and improving quality.
2. **Enabling Experimentation:** Collecting accurate performance data (often in real or near-real time) enables organizations to set up controlled experiments. This allows businesses to understand customer behavior and optimize services, such as when an online retailer compiles histories on customer clicks and interactions in real time.

3. **Replacing/Supporting Human Decision Making with Automated Algorithms:** Sophisticated **analytics** improve decision making, minimize risks, and unearth valuable insights. For example, retailers use algorithms to fine-tune inventories and pricing in response to real-time sales.
4. **Innovating New Business Models, Products, and Services:** Big Data allows companies to create new offerings, such as developing location-based services (like navigation or tailored insurance pricing) based on real-time location data.
5. **Analyzing Far More Data:** Organizations can process entire datasets rather than relying on sampling, which eliminates the risks associated with systematic biases in random sampling.
6. **Targeted Marketing and Customer Intimacy:** Big Data enables marketers to craft much more **personalized messages**. **Data mining**, a BI application that searches large datasets, helps predict trends and behaviors, allowing organizations to identify customer groups most likely to respond favorably to future marketing campaigns. This supports the strategic response of achieving **customer intimacy**.

List down the types of support provided by Accounting IS, finance IS, production/operations management (POM) IS, marketing IS, and human resources IS.

Functional Area Information Systems (FAISs) support a particular functional area within the organization by increasing each area's internal efficiency and effectiveness.

- *Accounting and Finance Information Systems (of money.
- **Financial Transaction Management:** Includes managing global stock exchanges, managing multiple currencies, performing a **virtual close** (closing books quickly to provide almost real-time financial health information), and

**expense (cost of money), managing financial transactions, managing cash, assessing financial health, and payroll.

Marketing Information Systems (MKT):

Successful organizations use IT to understand customers' needs and wants and develop marketing and advertising strategies around them.

- **Product Analysis:** Developing new goods and services.
- **Site Analysis:** Determining the best location for production and distribution facilities.
- **Promotion Analysis:** Identifying the best advertising channels.
- **Price Analysis:** Setting product prices to obtain the highest total revenues.
- **Customer Relations:** Used to know who customers are and treat them well.
- **Sales Force Automation:** Using software to automate the business tasks of sales, thereby improving the productivity of salespeople.

Production/Operations Management Information Systems (POM):

The POM function is responsible for processes that transform inputs into useful outputs.

- **Order Processing:** Managers use IT to process customer orders.
- **Production Scheduling:** Developing production schedules.
- **Inventory Control:** Controlling inventory levels, determining when and how much inventory to order, and how much to keep in stock.
- **Quality Control:** Monitoring product quality.
- **Product Design and Manufacturing:** Employing **computer-assisted design (CAD)** and **computer-assisted manufacturing (CAM)**.
- **Planning Production and Operations:** This has evolved from **Material Requirements Planning (MRP)** to **Manufacturing Resource Planning (MRP II)**.
- **Computer-Integrated Manufacturing (CIM):** An information system that integrates various automated factory systems.

Human Resources Information Systems (HRM):

HRIS applications have moved to intranets and the Web to perform key functions.

- **Recruitment:** Managing the recruiting process, analyzing, screening job applicants, and hiring new employees.
- **Maintenance and Development:** Helping employees manage their careers, administering performance tests, and monitoring employee productivity.
- **Compensation and Benefits:** Managing compensation and benefits packages.
- **Employee Records:** Maintaining employee records.
- **Employee Relationship Management (ERM):** Developing applications (like call centers) to better manage employees.

Analyse the main reasons of Computer Crimes.

Computer crimes are **illegal activities conducted over computer networks, particularly the Internet**. Modern cybercriminals are often part of well-organized criminal organizations that target known software security weaknesses for profit. Losses from computer crimes average hundreds of thousands of dollars, making them highly lucrative.

The increasing vulnerability of information resources, which makes computer crime possible, is driven by five key factors:

1. **Today's interconnected, interdependent, wirelessly networked business environment:** The evolution of IT infrastructure from mainframes to complex, wireless networks exposes organizations to a world of **untrusted networks** and potential attackers. Furthermore, **wireless is an inherently nonsecure broadcast communications medium**.
2. **Smaller, faster, cheaper computers and storage devices:** Devices such as netbooks, thumb drives, and iPads are becoming easier to steal and easier for attackers to use to steal information.
3. **Decreasing skills necessary to be a computer hacker:** The Internet contains computer programs called **scripts** that users with few skills can download and use to attack any information system connected to the Internet.

4. **International organized crime taking over cybercrime:** Groups of well-organized criminal organizations have taken control of a global billion-dollar crime network, powered by skillful hackers. Because computer crimes can be committed from anywhere, this provides an international safe haven for cybercriminals.
5. **Lack of management support:** Short-sighted management might reduce spending on information security countermeasures if no successful attacks occurred the previous year, leaving the company exposed.

These factors increase the likelihood of **deliberate threats**, which include software attacks (like viruses and worms), espionage or trespass, information extortion, identity theft, and cyberterrorism.

List the various components of CBIS.

A **computer-based information system (CBIS)** is an information system that uses computer technology to perform some or all of its intended tasks. The basic components of CBIS are categorized as follows:

Information Technology (IT) Components (which form the Information Technology Platform):

1. **Hardware:** Consists of devices such as the processor, monitor, keyboard, and printer. These accept, process, and display data and information.
2. **Software:** Is a program or collection of programs that enable the hardware to process data.
3. **Database:** Is a collection of related files or tables containing data.
4. **Network:** Is a connecting system (wireline or wireless) that permits different computers to share resources.

Other Components:

1. **Procedures:** Are the instructions for combining the above components to process information and generate the desired output.
2. **People:** Are those individuals who use the hardware and software, interface with it, or utilize its output.

IT components plus IT services (developing systems, managing security and risk, and managing data) comprise the organization's **information technology infrastructure**.

What are the features of Executive Support System?

The sources use the term **Dashboards** (or **digital dashboards**) to describe the system that supports managers across the organization. Dashboards tailored to the information needs of executives are called **executive dashboards**.

The key features of executive support systems/dashboards are:

- **Support for All Managers:** They are a special form of IS that support **all managers** of the organization.
- **Rapid Access to Timely Information:** They provide quick access to current data.
- **Direct Access to Structured Information:** They offer information in the form of reports.
- **Summarized Information:** They present structured, **summarized information** about aspects of business important to executives.
- **Top-level Placement:** They are located at the top of the hierarchy of information systems in an organization, supporting high-level decision-making.
- **Example:** A dashboard might show the **status of sales by product**.

Define Information security with an example.

Security can be defined as the degree of protection against criminal activity, danger, damage, and/or loss.

Information security refers to **all of the processes and policies designed to protect an organization's information and information systems (IS) from**

unauthorized access, use, disclosure, disruption, modification, or destruction.

Example:

After **Sony's PlayStation account database was hacked in 2011**, the company suffered a major information security breach, specifically related to unauthorized access and information theft. The company then had to pay **\$171 million** to rebuild its network and protect users from identity theft. This cost highlights the need for robust information security measures to protect critical assets like customer data and reputation.

Are Blogs and Wikis different? Justify with application of each.

Yes, **blogs and wikis are different**. Both are Web 2.0 tools used in social computing.

Tool	Definition/Characteristics	Application/Use	Source
Blog (Weblog)	An informal, personal journal that is frequently updated and is intended for general public reading.	Used in communications networks to provide fast, inexpensive communication. Marketers scan blogs to find information about potential employees. Used in collaborative CRM systems to enable customers to provide direct feedback.	
Wiki	A Web site on which anyone can post material and make changes to already posted material . They harness collective intelligence .	Used in collaboration tools to allow mutual efforts by two or more entities. Used in collaborative CRM systems to integrate customer feedback on	

Tool	Definition/Characteristics	Application/Use	Source
		marketing, sales, and support processes.	

Justification: While both are collaborative and social tools, **a blog is centered around a single or defined group authoring content** (like an online journal or commentary), while **a wiki is fundamentally designed for democratic, group creation, and editing of content.**

Highlight the Economic impacts of IS. Give example.

Information systems and information technology (IT) have profound economic impacts by altering costs, productivity, and market structures.

Economic Impacts Highlighted by IT:

- **Cost Management and Shift:** Cloud computing, which is built upon IS infrastructure, allows organizations to shift IT expenditures from **capital expenses (fixed costs)**, such as buying hardware and licenses, to **operational expenses (variable costs)**, where they pay only for the computing resources they need.
- **Productivity and Efficiency:** IT is instrumental in making businesses more productive, efficient, and responsive to consumers. CBIS performs high-speed, high-volume numerical computation and automates both semi-automatic business processes and manual tasks.
- **Labor Market Transformation:** IT may lead to the **elimination of jobs** or the **reduction of middle managers**. The manner in which displaced workers are treated raises ethical issues regarding compensation and retraining.
- **Globalization and Competition:** IT facilitates globalization, enabling more people to communicate, collaborate, and compete, thereby **leveling the digital playing field**. Having an Internet strategy is necessary for survival.

Example: Impact on Financial Markets

Global stock exchanges demonstrate a massive economic impact, as financial markets operate 24/7/365 using distributed electronic stock exchanges and the

Internet to buy and sell stocks and broadcast real-time prices. This instantaneous IT support, however, can introduce massive volatility. For instance, the **Flash Crash** incident (where automated high-frequency traders caused extreme market volatility) illustrated the disruptive, large-scale economic power of IT systems in global finance.

What do you mean by CAAS, SAAS, IASS? Give the application of each of these.

The provided sources define three cloud computing service models: IaaS, PaaS, and SaaS. The acronym **CAAS is not defined** in the source material.

These models represent the three types of computing generally required by consumers: infrastructure, platforms to develop applications, and software applications.

Model	Full Name	Definition	Application/Use Example	Source
IaaS	Infrastructure-as-a-Service	Cloud providers offer remotely accessible servers, networks, and storage capacity . Customers rent these computing resources.	Customers control the operating system and installed applications. Providers include Amazon, IBM, and Google.	
PaaS	Platform-as-a-Service	Customers rent servers, operating systems, storage, a database, and software development technologies (e.g., Java and	Allows customers to develop, run, and manage applications without the complexity of managing the infrastructure . Providers include Microsoft	

Model	Full Name	Definition	Application/Use Example	Source
		.NET) over the Internet.	Windows Azure and Force.com.	
SaaS	Software-as-a-Service	Vendors provide software specific to customers' requirements , hosting the applications and providing them as a service over a network (typically the Internet).	Allows applications to be accessible from various client devices through a thin-client interface . Flextronics used Workday (SaaS) for HR computing functions to save millions of dollars.	

Contrast to bring out the advantages and disadvantages of Compleitive environment in an organization.

The competitive environment is primarily analyzed using **Porter's Competitive Forces Model**, which recognizes five major forces that endanger a company's position. Advantages are gained when organizations successfully respond to these forces, often by implementing **Strategic Information Systems (SISs)**.

Competitive Force (Disadvantage/Threat)	Organizational Response (Advantage Gained via IS)	Source
Bargaining Power of Customers (Buyers): The Web increases buyer power by offering incredible amounts of choices and information.	Companies can implement loyalty programs in which they use the Web to monitor the activities of millions of customers, thereby reducing buyer power. Customer Relationship Management (CRM) is an effective IT response.	

Competitive Force (Disadvantage/Threat)	Organizational Response (Advantage Gained via IS)	Source
Threat of Substitute Products or Services: New technologies create substitutes rapidly (e.g., digitized media), and the Web makes information about substitutes available instantly.	The Web enables a company to build in switching costs , making it time-consuming and costly for customers to switch to a competitor.	
Rivalry Among Existing Firms in the Industry: Competition is highly intense; IT applications on the Web make proprietary systems visible and thus difficult to keep secret.	Organizations implement Strategic Information Systems (SISs) to gain competitive advantages such as increasing market share or profits. Differentiation and Cost Leadership strategies are employed.	
Threat of New Entrants: If entry barriers are low, new companies can easily enter the market.	Companies can raise entry barriers by offering product or service features that customers expect (e.g., high levels of customer service or unique technological features).	
Bargaining Power of Suppliers: Suppliers gain power when they are unique or limited.	Organizations can use IT to implement a buy-side marketplace (reverse auctions) to find cheaper sources and reduce supplier leverage.	

Discuss how privacy issue can impact transborder data flows?

Information privacy is the right to determine when, and to what extent, personal information can be gathered and/or communicated to others. When data crosses international borders (**transborder data flows**), privacy issues become legally complex because data regulations differ among countries.

Privacy concerns impact transborder data flows in the following ways:

- **Jurisdictional Complexity:** When utilizing cloud computing, customer data may not remain in the same data center or even the same country, complicating data privacy and raising concerns over **legal jurisdiction**.
- **EU Data Transfer Restrictions:** The **European Union explicitly prohibits consumers' data from being transferred to nonmember countries** without the consumers' prior consent and approval. This severely restricts how businesses outside the EU can process EU citizen data.
- **Safe Harbor Requirement:** Companies outside the EU can overcome this restriction only by demonstrating that they provide a **"safe harbor"** for the data.
- **Compliance Costs and Constraints:** To achieve compliance with stringent data protection laws (such as the EU Data Protection Directive or Germany's restrictive data export laws), cloud computing customers often have to adopt **hybrid deployment modes**. These modes are typically more expensive and may offer restricted benefits compared to a fully open public cloud model.

What are types IS? Explain with example.

Information Systems (IS) collect, process, store, analyze, and disseminate information for a specific purpose. Modern organizations employ many different types of Computer-Based Information Systems (CBISs).

Type of System	Function	Example	Source
Transaction Processing System (TPS)	Processes transaction data from business events.	Walmart checkout point-of-sale terminal.	
Functional Area IS (FAIS)	Supports the activities within a specific functional area.	System for processing payroll.	
Enterprise Resource Planning (ERP)	Integrates all functional areas of the organization.	Oracle, SAP system.	
Office Automation System (OAS)	Supports daily work activities of individuals and groups.	Microsoft® Office.	

Type of System	Function	Example	Source
Business Intelligence (BI) System	Provides computer-based support for complex, nonroutine decisions.	"What-if" analysis of changes in budget.	
Expert System (ES)	Mimics human expert in a particular area and makes decisions.	Credit card approval analysis.	
Executive Dashboard	Presents structured, summarized information about aspects of business important to executives.	Status of sales by product.	
Supply Chain Management (SCM) System	Manages flows of products, services, and information among organizations.	Walmart Retail Link system connecting suppliers to Walmart.	
Electronic Commerce (E-commerce) System	Enables transactions among organizations and between organizations and customers.	www.dell.com .	

Identify advantages and drawbacks of businesses implementation which uses an enterprise resource planning system.

Enterprise Resource Planning (ERP) systems are information systems that integrate the planning, management, and use of all of an organization's resources by taking a business process view of the overall organization, employing a common software platform and database.

Advantages (Benefits) of ERP Systems:

1. **Flexibility and Adaptability:** Because ERP systems integrate organizational resources, they make organizations more **flexible, agile, and adaptive**. They enable quick reactions to changing business conditions and allow capitalizing on new business opportunities.

2. **Improved Decision Making:** ERP systems provide essential information on business performance across functional areas, which significantly improves managers' ability to make **better, more timely decisions**.
3. **Quality Improvement:** Integration results in significant improvements in the **quality of customer service, production, and distribution**.
4. **Integration of Functional Areas:** The major objective is to tightly integrate the functional areas of the organization so information flows seamlessly.

Drawbacks (Disadvantages) of ERP Systems:

1. **High Cost:** ERP systems can be **extremely expensive** to acquire, implement, and maintain.
2. **Process Change Requirement:** Implementation often forces organizations to change their existing business processes to fit the predefined business processes incorporated into the ERP software.
3. **Complexity and Difficult Integration:** ERP systems are inherently complex, and integrating them with other legacy organizational systems can be difficult.
4. **Cloud-Specific Risks:** If implemented using a cloud-based model, companies might:
 - Face **security concerns** (the primary reason organizations avoid cloud ERP).
 - **Sacrifice control** over a strategic IT resource.
 - Encounter problems that are difficult to fix due to the vendor managing the system.

Briefly describe the risks of social computing to business giving suitable examples.

Social computing is a type of IT that combines social behavior and information systems, focusing on collaboration, interaction, and encouraging user-generated

content. Because the content is often user-generated and unfiltered, it poses several risks to businesses.

The risks associated with social computing include:

1. **Information Security and Data Leakage:** Concerns include data leakage of personal information or corporate strategic information. For example, employees might disclose confidential information accidentally.
2. **Privacy and Intellectual Property Violations:** Invasion of privacy and violation of intellectual property and copyright are constant risks.
3. **Reputation Damage from Unfiltered Content:** Businesses must be willing to accept negative reviews and feedback, as poor or biased quality of user-generated content can spread rapidly through social networks.
4. **Employee Behavior Risks:** Issues include cyberbullying, cyberstalking, and employee harassment. Furthermore, employees' reluctance to participate in corporate social initiatives can diminish their effectiveness.

Example:

To mitigate the risk of negative feedback and biased content, companies must monitor their social presence. **Rosetta Stone**, which sells language software, implemented a strategy involving both human intervention and software to help monitor Wall posts on Facebook and respond to them constructively. This demonstrates a business response to the risk posed by unfiltered user feedback.

Is security an ethical responsibility? Justify with a case study.

Yes, **security is inherently an ethical responsibility** because the principles of ethics demand accountability for actions taken concerning information assets.

Justification:

1. **Ethical Foundation (Responsibility and Accountability):** **Ethics** refers to the principles of right and wrong used to guide behavior. The fundamental tenets of ethics include **responsibility** (accepting consequences of decisions) and **accountability** (determining who is responsible for actions). Since IT security policies determine how information is collected, stored, and protected, failing

to implement adequate security represents a failure of ethical responsibility, particularly when it leads to damage or loss.

2. **Protecting Ethical Issues:** Information security directly addresses the four major categories of ethical issues related to IT: **privacy, accuracy, property, and accessibility**. Breaches compromise customers' privacy and jeopardize the integrity and ownership of organizational data.
3. **MIS Role:** MIS personnel, who are responsible for the security infrastructure and control the organization's information assets and vast amounts of personal employee information, must be held to the highest ethical standards.

Case Study Justification (Target):

The closing case regarding **Target's data collection and analysis** illustrates the ethical responsibility tied to security. Target maintains massive amounts of customer data (purchases, preferences, products). While legal, the collection and analysis of such detailed personal information raise significant ethical questions about **privacy invasion**. If Target's security failed, the breach would not only be a legal and financial disaster but also a failure of the ethical responsibility to protect the sensitive information entrusted to them. Managers must ensure security to maintain trust and protect the privacy rights of their customers.

Analyze the key benefits of cloud computing

Cloud computing is a type of computing that delivers convenient, on-demand, pay-as-you-go access to a shared pool of configurable computing resources (servers, networks, storage, applications, and services) that can be rapidly and easily accessed over the Internet.

The key benefits derived from cloud computing include:

1. **Cost Reduction and Flexibility (Fixed to Variable Costs):** Cloud computing helps reduce fixed IT costs by enabling organizations to shift from **capital expenses** (buying licenses, servers) to **operational expenses** (paying for services as needed). This pay-for-use model provides greater flexibility. Example: Amazon Web Services reduced its prices a dozen times between 2008 and 2012.

2. **Business Scalability and Adaptability:** Cloud computing allows organizations to use the precise amount of computing resources needed, enabling them to quickly increase the scope of their business operations and react rapidly to **fluctuating and unpredictable business demand**.
3. **Improved Utilization of Hardware/Infrastructure:** Cloud computing eases the difficult tasks of procuring, configuring, and maintaining hardware and software, freeing up internal IT resources. Resources are shared and pooled, improving utilization.
4. **Faster Application Development and Deployment:** It allows enterprises to get their applications up and running faster, with easier manageability and less maintenance.
5. **Enhanced Collaboration:** Cloud computing facilitates collaboration, allowing geographically distributed development teams (or organizations) to work together on software projects.
6. **Data Insight:** Cloud computing supports the use of analytics tools for gathering, storing, accessing, and analyzing vast amounts of data to mine insights.
7. **Increased Individual Productivity:** Cloud computing generally makes individuals more productive.

How the quality of data is ensured in an organization?

Ensuring data quality is a critical aspect of **data governance**, which is an approach to managing information across an entire organization to ensure data are handled in a well-defined fashion. Poor data quality can result in data that are not trusted and ultimately unused.

Data quality is ensured through processes and standards:

1. **Master Data Management (MDM):** This strategy is a process that provides companies with the ability to store, maintain, exchange, and synchronize a consistent, accurate, and timely **"single version of the truth"** for the company's core **master data** (e.g., customer, product, employee data).

2. **Normalization:** This is a method used in relational databases for analyzing and reducing the database to its most streamlined form for minimum redundancy and **maximum data integrity**. **Data integrity** means the data meets certain constraints (e.g., no alphabetic characters in a Social Security number field).
3. **Improvement at the Source:** Although data quality can sometimes be improved with data-cleansing software, the better, long-term solution is to **improve the quality at the source system level**. This requires the **business owners of the data to assume responsibility** for making any necessary changes.
4. **Metadata:** Maintaining **data about the data** (metadata) in a data warehouse is important for both IT personnel and users to ensure clarity regarding data definitions, sources, and accuracy.
5. **Auditing:** Information systems auditing procedures examine the integrity, security, and proper functioning of systems and their inputs and outputs to ensure all controls are installed and effective.